

## QueryBook — Drawings, Reference Numerals & Brief Description of the Drawings

**Scope:** the complete, current QueryBook figure set — FIG. 1–FIG. 49 — as one unified, contiguously numbered set, with no gaps. It comprises: the integral language & speech subsystem (LEL; FIG. 1–8 and FIG. 31–32); the QueryBook cognitive platform — Prime Directive / Domain 0, the Fact Unit, UFCS, FQL, the neuromorphic cognitive layer, the user-owned agent system, the enhanced TCP/IP hybrid protocol, and the media pipeline (FIG. 9–30); the Language & Identity Linking (LIL) voiceprint subsystem (FIG. 33–34); the Universal Integrity Audit Service (UIAS; FIG. 35–36); the Paralinguistic & Pragmatic Interpretation Layer (PIL; FIG. 37–38); Lecture Query mode (FIG. 39–40); and the recent extensions — deterministic multilingual acquisition, Dialect Parameter Clusters, grounded scene compilation, key-safe external-service mediation, scene reconstruction, modular editions, and the version-independent provenance-continuous store (FIG. 41–47); and the application embodiments — the cited Q&A / ChatBot application and the application-and-deployment surfaces (on-device eBook/document Q&A, SaaS/API, the Fact-Unit compressed search engine, the deterministic LLM-hybrid adjunct, and private/offline analysis) over the invariant core (FIG. 48–49). **Companion to** the consolidated *QueryBook Provisional Patent Application*; the figures, reference numerals, and brief descriptions here are identical to those in that application. **Document date:** October 7, 2026. Draft — to be finalized to USPTO drawing standards (37 CFR 1.84) by a patent draftsman before filing.

# Feature Inventory and Figure Map

This inventory lists the major QueryBook features and the figure that depicts each, confirming that every major feature — the cognitive core (Fact Units, UFCS, FQL, the Prime Directive, the neuromorphic layer), the agent and ChatBot applications, security, media, language and speech, and the identity / integrity / interpretation / lecture subsystems — is drawn. **Status** is conservative: "Prototype" marks a capability exercised in the current prototype; "Specified" marks a described embodiment; the semantic media pipeline of FIG. 24 is identified as specified and not yet built or measured. 49 features are mapped across FIG. 1–FIG. 49.

| Feature                                                                                   | Figure         | Status                                |
|-------------------------------------------------------------------------------------------|----------------|---------------------------------------|
| <b>A. Primitive and substrate</b>                                                         |                |                                       |
| Fact Unit — atomic subject-predicate-object-polarity with content-derived identity (FUID) | <b>FIG. 10</b> | Prototype                             |
| Fact Unit confidence and evidence model (Bayesian / Beta)                                 | <b>FIG. 27</b> | Prototype                             |
| Fact-certification authority hierarchy                                                    | <b>FIG. 11</b> | Specified                             |
| UFCS — Universal Federated Connectivity Substrate (three-store)                           | <b>FIG. 12</b> | Prototype                             |
| UFCS packet structure — serialized Fact Unit record                                       | <b>FIG. 28</b> | Specified                             |
| UFCS content-addressed block framing                                                      | <b>FIG. 29</b> | Specified                             |
| <b>B. Query, reasoning and governance core</b>                                            |                |                                       |
| Fifteen-domain architecture; Prime Directive (Domain 0)                                   | <b>FIG. 9</b>  | Prototype                             |
| FQL — Federated Query Language: joint structure + trust execution                         | <b>FIG. 14</b> | Prototype                             |
| FQL — query anatomy and worked example (cite or refuse)                                   | <b>FIG. 30</b> | Prototype                             |
| Neuromorphic cognitive layer — attractor convergence                                      | <b>FIG. 26</b> | Specified                             |
| Adaptive response engine — directive fixed before generation                              | <b>FIG. 21</b> | Prototype                             |
| Verifying machine-generated text against certified records                                | <b>FIG. 15</b> | Specified                             |
| <b>C. Knowledge operations and deployment</b>                                             |                |                                       |
| Authored tour / exhibition sequencing                                                     | <b>FIG. 13</b> | Specified                             |
| Federation — cross-boundary re-validation                                                 | <b>FIG. 16</b> | Specified                             |
| Fact-level erasure by external-key destruction                                            | <b>FIG. 17</b> | Specified                             |
| User-owned Agent system with governance and kill switch                                   | <b>FIG. 22</b> | Specified                             |
| Deployment topologies over an invariant common layer                                      | <b>FIG. 20</b> | Specified                             |
| Modular capability editions (light / medium / full)                                       | <b>FIG. 46</b> | Prototype                             |
| Version-independent, provenance-continuous store                                          | <b>FIG. 47</b> | Prototype                             |
| <b>D. Security</b>                                                                        |                |                                       |
| Universal multi-layer security architecture                                               | <b>FIG. 18</b> | Specified                             |
| Universal honeypot saturation / data firehose (anti-Agent-Worm)                           | <b>FIG. 19</b> | Specified                             |
| <b>E. Media and protocol</b>                                                              |                |                                       |
| Deterministic multi-language regeneration                                                 | <b>FIG. 23</b> | Specified                             |
| Semantic media pipeline — compression · expansion · colorization · upscaling              | <b>FIG. 24</b> | Specified — not yet built or measured |
| Grounded scene compilation to an external generative media engine                         | <b>FIG. 43</b> | Prototype                             |
| Key-safe external-service mediation                                                       | <b>FIG. 44</b> | Prototype                             |
| Scene reconstruction from prose (silhouette-protected)                                    | <b>FIG. 45</b> | Prototype                             |
| QueryBook-enhanced TCP/IP hybrid protocol packet                                          | <b>FIG. 25</b> | Specified                             |
| <b>F. Language and speech (LEL)</b>                                                       |                |                                       |
| LEL overall system architecture                                                           | <b>FIG. 1</b>  | Specified                             |
| Sub-Language Priming Layer (SLPL)                                                         | <b>FIG. 2</b>  | Specified                             |
| Developmental learning sequence and one-way transition gate                               | <b>FIG. 3</b>  | Prototype                             |
| Multilingual delta acquisition and acceleration flywheel                                  | <b>FIG. 4</b>  | Specified                             |
| Speech output pipeline (seven learned stages)                                             | <b>FIG. 5</b>  | Specified                             |
| Co-grounded integration (LEL + FQL + UFCS)                                                | <b>FIG. 6</b>  | Specified                             |
| Language Lab reduced-to-practice prototype (Phases 1-2)                                   | <b>FIG. 7</b>  | Prototype                             |
| Deterministic Phase-2 semantic grounding (dictionary + store, no LLM)                     | <b>FIG. 8</b>  | Prototype                             |
| Advanced multilingual pronunciation and self-contained speech                             | <b>FIG. 31</b> | Prototype                             |
| Permanent pronunciation / speech quality-control battery                                  | <b>FIG. 32</b> | Prototype                             |
| Deterministic multilingual acquisition (no generative model)                              | <b>FIG. 41</b> | Prototype                             |
| Dialect Parameter Clusters (DPC)                                                          | <b>FIG. 42</b> | Prototype                             |
| <b>G. Identity, integrity, interpretation and lectures</b>                                |                |                                       |
| LIL — voiceprint enrollment and matching over the identity graph                          | <b>FIG. 33</b> | Prototype                             |
| LIL — trust-gated voice authentication with anti-spoofing                                 | <b>FIG. 34</b> | Specified                             |
| UIAS — pre-dispute sealing and cross-submission similarity                                | <b>FIG. 35</b> | Specified                             |
| UIAS — signed Proof Package generation                                                    | <b>FIG. 36</b> | Specified                             |
| PIL — paralinguistic and pragmatic interpretation of an ingested voice                    | <b>FIG. 37</b> | Prototype                             |
| PIL — human query interpretation (clarify-or-answer)                                      | <b>FIG. 38</b> | Prototype                             |
| Lecture Query — ingestion pipeline                                                        | <b>FIG. 39</b> | Prototype                             |
| Lecture Query — forever-linked record and its querying                                    | <b>FIG. 40</b> | Prototype                             |
| <b>H. Application embodiments</b>                                                         |                |                                       |
| Cited Q&A / ChatBot application (citation-or-UNKNOWN end to end)                          | <b>FIG. 48</b> | Prototype                             |
| Application and deployment surfaces over the invariant core                               | <b>FIG. 49</b> | Prototype                             |

## Brief Description of the Drawings

**FIG. 1** is a block diagram of the overall QueryBook Language Expression Layer (LEL) system architecture.

**FIG. 2** is a block diagram of the Sub-Language Priming Layer (SLPL) and the measurable representation produced by each of its six components.

**FIG. 3** is a flow diagram of the developmental learning sequence and the one-way Transition Gate having four readiness thresholds.

**FIG. 4** is a block diagram of the multilingual Delta Acquisition Model and the acquisition acceleration flywheel.

**FIG. 5** is a block diagram of the seven-stage Speech Output Pipeline.

**FIG. 6** is a data-flow diagram of the co-grounded integration of the LEL with the Federated Query Language (FQL) and the Universal Federated Connectivity System (UFCS).

**FIG. 7** is a block diagram of a reduced-to-practice embodiment (the Language Lab) implementing Phase One, the Transition Gate, and a deterministic Phase Two, per Section VII.

**FIG. 8** is a data-flow diagram of the deterministic Phase-Two semantic grounding of the reduced-to-practice embodiment, in which word meanings are attached only from auditable sources — a bundled public-domain dictionary and self-grounding against verified Fact Units in the store — with any large-language-model meaning source excluded from the grounding path.

## Brief Description of the QueryBook Platform Drawings (FIG. 9–FIG. 30)

**FIG. 9** is a diagram of the queryBook system architecture (Prime Directive as Domain 0).

**FIG. 10** is a diagram of the fact Unit — atom, envelope, and content-derived identity.

**FIG. 11** is a diagram of the fact-certification authority hierarchy.

**FIG. 12** is a diagram of the uFCS substrate — three-store architecture.

**FIG. 13** is a diagram of the authored tour and exhibition sequencing.

**FIG. 14** is a diagram of the fQL — joint structure and trust execution.

**FIG. 15** is a diagram of the verifying machine-generated text against certified records.

**FIG. 16** is a diagram of the federation — cross-boundary re-validation.

**FIG. 17** is a diagram of the fact-level erasure by external-key destruction.

**FIG. 18** is a diagram of the universal security architecture.

**FIG. 19** is a diagram of the universal honeypot saturation and data firehose.

**FIG. 20** is a diagram of the deployment topologies over an invariant common layer.

**FIG. 21** is a diagram of the adaptive response engine — directive fixed before generation.

**FIG. 22** is a diagram of the user-owned agent system with governance and kill switch.

**FIG. 23** is a diagram of the deterministic multi-language regeneration.

**FIG. 24** is a diagram of the semantic compression, expansion, colorization and upscaling.

**FIG. 25** is a diagram of the queryBook-enhanced TCP/IP hybrid protocol packet.

**FIG. 26** is a diagram of the neuromorphic cognitive layer — attractor convergence.

**FIG. 27** is a diagram of the fact Unit — confidence and evidence model.

**FIG. 28** is a diagram of the uFCS packet structure — serialized Fact Unit record.

**FIG. 29** is a diagram of the uFCS packet framing — content-addressed block storage.

**FIG. 30** is a diagram of the fQL — query anatomy and worked example.

## Reference Numerals

| No. | Element |
|-----|---------|
|-----|---------|

|     |                                                |
|-----|------------------------------------------------|
| 100 | QueryBook LEL system                           |
| 110 | Sub-Language Priming Layer (SLPL)              |
| 120 | Language Expression Layer (LEL)                |
| 130 | Transition Gate                                |
| 140 | Phase Two semantic-communicative learning      |
| 150 | Multilingual Delta Acquisition module          |
| 160 | Speech Output Pipeline                         |
| 165 | natural speech output                          |
| 170 | co-grounding interface                         |
| 180 | Federated Query Language (FQL)                 |
| 190 | Universal Federated Connectivity System (UFCS) |
| 195 | federated data sources                         |
| 200 | SLPL                                           |
| 210 | acoustic phonetic processing                   |
| 211 | signal-level feature embedding                 |
| 220 | phonics / phonemic awareness                   |
| 221 | phoneme-boundary segmenter                     |
| 230 | communication-science foundations              |
| 231 | communicative-salience weight                  |
| 240 | prosody & rhythm processing                    |
| 241 | prosodic contour vector                        |
| 250 | information-theoretic grounding                |
| 251 | information weighting                          |
| 260 | neurolinguistic chunking                       |
| 261 | chunk segments                                 |
| 270 | licensed multimedia input                      |
| 300 | developmental learning sequence                |
| 310 | Phase One (structural-acoustic)                |
| 311 | sound discrimination                           |
| 312 | syllabic segmentation                          |
| 313 | word-as-acoustic-object                        |
| 314 | semantics suppressed                           |
| 320 | Transition Gate (one-way)                      |
| 321 | phonemic-inventory completeness $\geq \tau_1$  |
| 322 | lexical stability $\geq \tau_2$                |
| 323 | co-occurrence density $\geq \tau_3$            |
| 324 | pattern saturation $< \epsilon$                |
| 330 | Phase Two (semantic-communicative)             |
| 331 | word-to-referent grounding                     |
| 332 | communicative discovery                        |
| 333 | emergent grammar / syntax                      |
| 400 | Delta Acquisition Model                        |
| 410 | reused L1 foundation                           |
| 420 | L2 delta detector                              |
| 430 | targeted training of novel elements            |
| 440 | multilingual semantic alignment                |
| 450 | acquisition acceleration flywheel              |
| 451 | negative-transfer detection & retraining       |
| 500 | Speech Output Pipeline                         |
| 510 | lexical selection                              |
| 520 | grapheme-to-phoneme conversion                 |
| 530 | coarticulation modeling                        |
| 540 | prosody generation                             |
| 550 | register & affect calibration                  |
| 560 | self-monitoring feedback loop                  |
| 570 | neural vocoder rendering                       |

| No. | Element |
|-----|---------|
|-----|---------|

|      |                                                                            |
|------|----------------------------------------------------------------------------|
| 984  | engine-selection ladder (per OS / language)                                |
| 985  | host OS built-in voice service                                             |
| 986  | culture-matched OS voice (when installed)                                  |
| 987  | honest refusal with remedy (no fabricated audio)                           |
| 990  | deterministic translator (through English)                                 |
| 991  | bundled open-source bilingual dictionary                                   |
| 992  | coverage accounting + unknown-word passthrough (flagged)                   |
| 993  | deterministic language detector                                            |
| 995  | permanent quality-control battery                                          |
| 996  | IPA-produced check                                                         |
| 997  | syllable-sanity check ( $\pm 1$ )                                          |
| 998  | valid-waveform synthesis check                                             |
| 999  | per-language QC report + overall pass score                                |
| 1000 | LIL voiceprint & identity subsystem                                        |
| 1001 | consent gate (precondition)                                                |
| 1002 | audio/video capture + VAD/segmentation                                     |
| 1003 | acoustic feature extraction                                                |
| 1004 | speaker embedding (voiceprint vector)                                      |
| 1005 | enrollment — canonical voiceprint                                          |
| 1006 | match / score (cosine · PLDA)                                              |
| 1007 | QueryBook trust gate<br>(VERIFIED/CONTRADICTED/UNKNOWN)                    |
| 1008 | Fact-Unit identity graph                                                   |
| 1009 | LEL cross-script name/entity resolution                                    |
| 1010 | provenance-tracked link Fact Unit                                          |
| 1020 | voice authentication request                                               |
| 1021 | capture phrase + embed                                                     |
| 1022 | anti-spoofing (liveness/replay/synthetic)                                  |
| 1023 | deterministic risk-scoring policy                                          |
| 1024 | authentication decision gate                                               |
| 1025 | second-factor challenge                                                    |
| 1026 | auth-event Fact Unit (provenance)                                          |
| 1100 | UIAS integrity audit service                                               |
| 1101 | monitored system workflow (unmodified)                                     |
| 1102 | input-boundary position                                                    |
| 1103 | processing-layer position                                                  |
| 1104 | storage-layer position                                                     |
| 1105 | output-boundary / handoff position                                         |
| 1106 | Connector + Ingestion agents (seal at observation)                         |
| 1107 | seal = Fact Unit (hash · id-hash · independent timestamp · source · agent) |
| 1108 | Similarity Detection Agent (semantic signatures)                           |
| 1109 | Fact-Unit audit ledger (UFCS, append-only, immutable)                      |
| 1110 | contemporaneous cross-party anomaly flag                                   |
| 1111 | Verification Agent (continuous re-verification)                            |
| 1112 | content-not-stored / one-way identity (fact-level erasure)                 |
| 1120 | UIAS signed Proof Package generation                                       |
| 1121 | accusation / inquiry                                                       |
| 1122 | Proof Package Agent                                                        |
| 1123 | sealed Fact Units + verified seals                                         |
| 1124 | cross-submission similarity log entry                                      |
| 1125 | chronological ledger extract (RPH)                                         |
| 1126 | Arbitration Agent (factual determination; Prime-Directive gate)            |
| 1127 | assemble + sign under UIAS independent identity                            |
| 1128 | signed Proof Package                                                       |
| 1129 | factual-only determination (legal reserved to courts/regulators)           |
| 1200 | PIL paralinguistic & pragmatic interpretation                              |
| 1201 | ingested voice (consent-gated)                                             |

| No. | Element                                                                  | No.  | Element                                                             |
|-----|--------------------------------------------------------------------------|------|---------------------------------------------------------------------|
| 580 | natural speech waveform                                                  | 1202 | transcription (words)                                               |
| 600 | co-grounded integration                                                  | 1203 | Paralinguistic Analyzer (measured)                                  |
| 610 | natural-language input                                                   | 1204 | tone/loudness/pacing/emphasis + labeled arousal estimate            |
| 620 | LEL analysis                                                             | 1205 | Pragmatic Framer (sourced)                                          |
| 630 | FQL operations                                                           | 1206 | context · perspective · ergonomics · culture · social mores         |
| 640 | UFCS execution                                                           | 1207 | Interpretation Frame (labeled estimates + sourced frames)           |
| 650 | federated sources                                                        | 1208 | Fact Units on ingested content                                      |
| 660 | results                                                                  | 1209 | covenant: estimates not asserted; frames sourced; consent           |
| 670 | render to NL / speech in input language                                  | 1210 | PIL human query interpretation                                      |
| 680 | language-agnostic resolution                                             | 1211 | human query (typed/spoken)                                          |
| 700 | Language Lab prototype embodiment                                        | 1212 | Interpretation Frame (para + pragmatic)                             |
| 710 | corpus source                                                            | 1213 | intent disambiguation (context/perspective)                         |
| 720 | Phase-One structural agent                                               | 1214 | ambiguity gate                                                      |
| 730 | structural analyzer                                                      | 1215 | ask one clarifying question / stop                                  |
| 731 | grapheme inventory                                                       | 1216 | register & ergonomic mode selector                                  |
| 732 | rule-seeded G2P                                                          | 1217 | culture/mores/ergonomics packs                                      |
| 733 | syllabifier                                                              | 1218 | retrieve-gate-compose (verified Fact Units; cite or refuse)         |
| 734 | lexical-stability metric                                                 | 1219 | answer in selected register/mode (facts unchanged)                  |
| 735 | co-occurrence density metric                                             | 1300 | Lecture Query ingestion pipeline                                    |
| 736 | prosody proxy                                                            | 1301 | live or recorded audio/video program                                |
| 740 | Fact Unit store                                                          | 1302 | capture + consent gate                                              |
| 750 | Transition-Gate metrics                                                  | 1303 | ASR transcription + diarization (untrusted input)                   |
| 751 | one-way latch                                                            | 1304 | video frames → faces (LIL)                                          |
| 760 | phase agents on shared manager                                           | 1305 | PIL analysis over time (tone/pacing/arousal + frame)                |
| 761 | Phase-1 agent (implemented)                                              | 1306 | LEL segment/structure/extract claims                                |
| 762 | Phase-2 agent — deterministic dictionary + store grounding (implemented) | 1307 | deterministic summary (TOC/key points/claims)                       |
| 763 | Phase-3 agent (declines — spec-only)                                     | 1308 | ingest → Lecture Record; claims = attributed utterance facts        |
| 764 | Phase-4 agent (declines — spec-only)                                     | 1309 | forever-link: speaker/venue/event/org/date (immutable, trust-gated) |
| 800 | deterministic Phase-Two semantic grounding                               | 1310 | forever-linked Lecture Record (queryable)                           |
| 810 | Phase-One vocabulary (learned words)                                     | 1311 | Lecture Record (content-addressed)                                  |
| 820 | deterministic grounder (per word)                                        | 1312 | speaker entity (LIL voiceprint/face)                                |
| 830 | bundled public-domain dictionary source                                  | 1313 | venue entity                                                        |
| 831 | dictionary meaning Fact Unit (means, trust 0.9)                          | 1314 | event entity                                                        |
| 840 | self-grounding store-lookup source                                       | 1315 | organization + date                                                 |
| 841 | store-link Fact Unit (grounded_by_fact)                                  | 1316 | transcript + diarization                                            |
| 850 | Fact Unit store (provenance, domain language)                            | 1317 | PIL track (tone/pacing over time)                                   |
| 860 | excluded large-language-model meaning source                             | 1318 | attributed claims → corroborated/contradicted/open                  |
| 870 | optional future dictionary-gated suggestor                               | 1319 | FQL / Q&A over the record                                           |
| 900 | advanced multilingual pronunciation subsystem                            | 1320 | answer: cited, attributed, refused where unknown                    |
| 910 | pronunciation request (word/phrase, language)                            | 4800 | cited Q&A / ChatBot application                                     |
| 920 | pronunciation preference-ladder selector                                 | 4801 | user question (typed/spoken)                                        |
| 930 | bundled open-source phonemizer                                           | 4810 | query interpretation (intent/register)                              |
| 931 | phonemizer IPA transcription                                             | 4820 | FQL retrieval over verified Fact Units                              |
| 932 | phonemizer call cache                                                    | 4830 | Prime-Directive gate (evidence sufficiency)                         |
| 933 | bounded call log / timeout guard                                         | 4840 | composed answer with per-fact citations                             |
| 940 | rule-seeded G2P tables (per language)                                    | 4841 | honest UNKNOWN (refuse, never guess)                                |
| 941 | approximate rule-seeded IPA (offline)                                    | 4850 | ChatBot/assistant delivery surface                                  |
| 950 | syllable count + primary-stress derivation                               | 4860 | citation-or-UNKNOWN covenant held end to end                        |
| 960 | explicit null result (not fabricated)                                    | 4900 | invariant QueryBook core                                            |
| 970 | phrase-level single-call transcription                                   | 4910 | on-device eBook/document Q&A (micro-encapsulated)                   |
| 971 | displayed transcription consistent with synthesized audio                | 4920 | SaaS / API service surface                                          |
| 980 | self-contained application bundle                                        | 4930 | Fact-Unit compressed search engine                                  |
| 981 | bundled open-source speech engine                                        | 4940 | deterministic LLM-hybrid adjunct (gated suggestor)                  |
| 982 | runtime-support libraries                                                | 4950 | private/offline analysis deployment                                 |
| 983 | bundled phoneme data                                                     | 4960 | all surfaces inherit the governed-fact contract                     |

## Drawings

Formal patent drawings — black line art, forming **one QueryBook figure set**: FIG. 1–FIG. 8 and FIG. 31–FIG. 32 depict QueryBook’s integral developmental language and speech subsystem (the Language Expression Layer), FIG. 9–FIG. 30 depict the QueryBook platform in which that subsystem is co-grounded (see FIG. 6), FIG. 33–FIG. 34 depict the Language & Identity Linking (LIL) voiceprint and multimodal identity subsystem, FIG. 35–FIG. 36 depict the Universal Integrity Audit Service (UIAS) that QueryBook delivers on these primitives, FIG. 37–FIG. 38 depict the Paralinguistic & Pragmatic Interpretation Layer (PIL), and FIG. 39–FIG. 40 depict the Lecture Query interaction mode. Each figure is on its own sheet and every element is identified by a reference numeral used consistently in the Brief Description of the Drawings, the Reference Numerals key, and the Detailed Description. Figures are draft; counsel/draftsperson to finalize to USPTO drawing standards (37 CFR 1.84) before filing.

FIG. 1

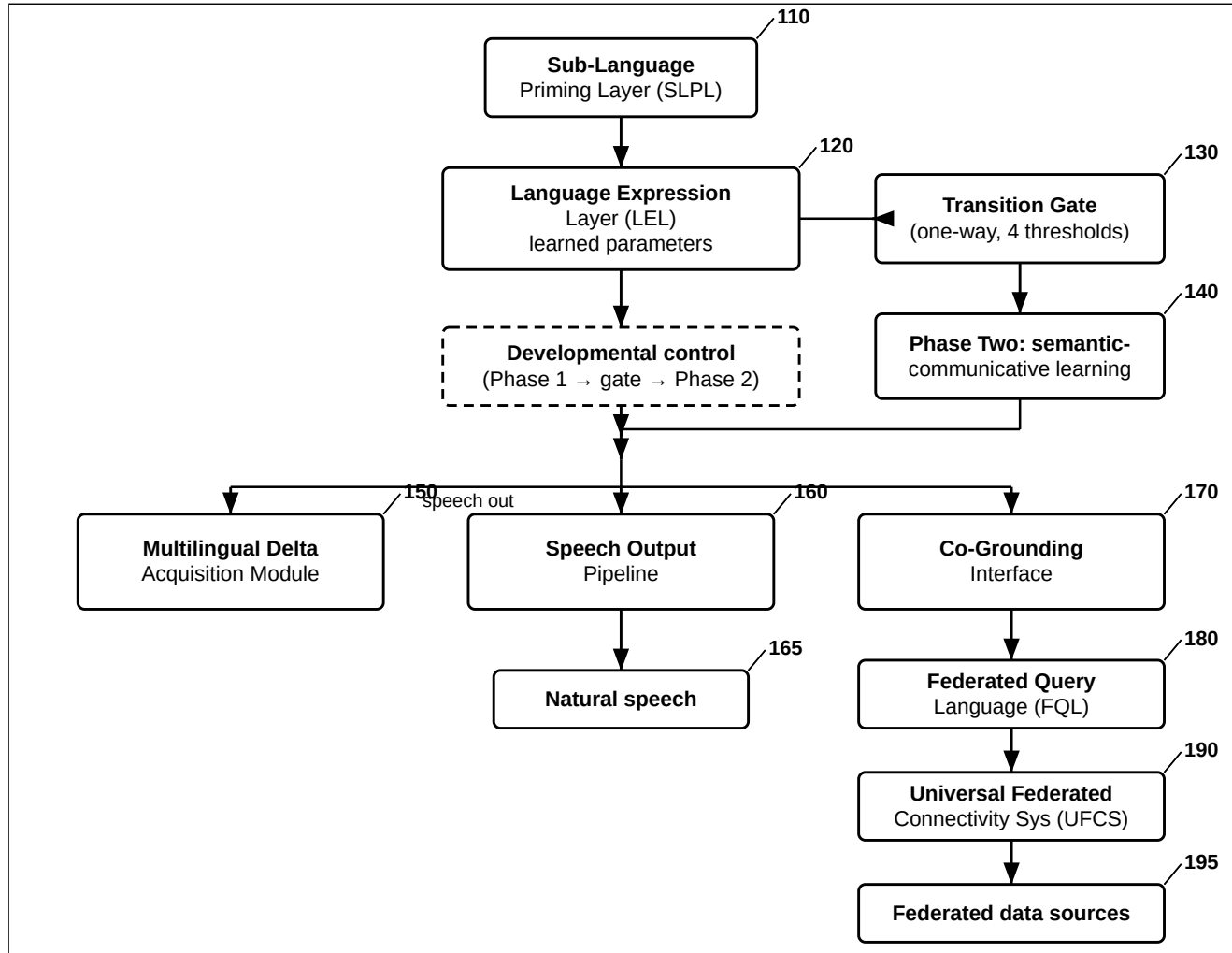


FIG. 1 — QueryBook LEL — Overall System Architecture

FIG. 2

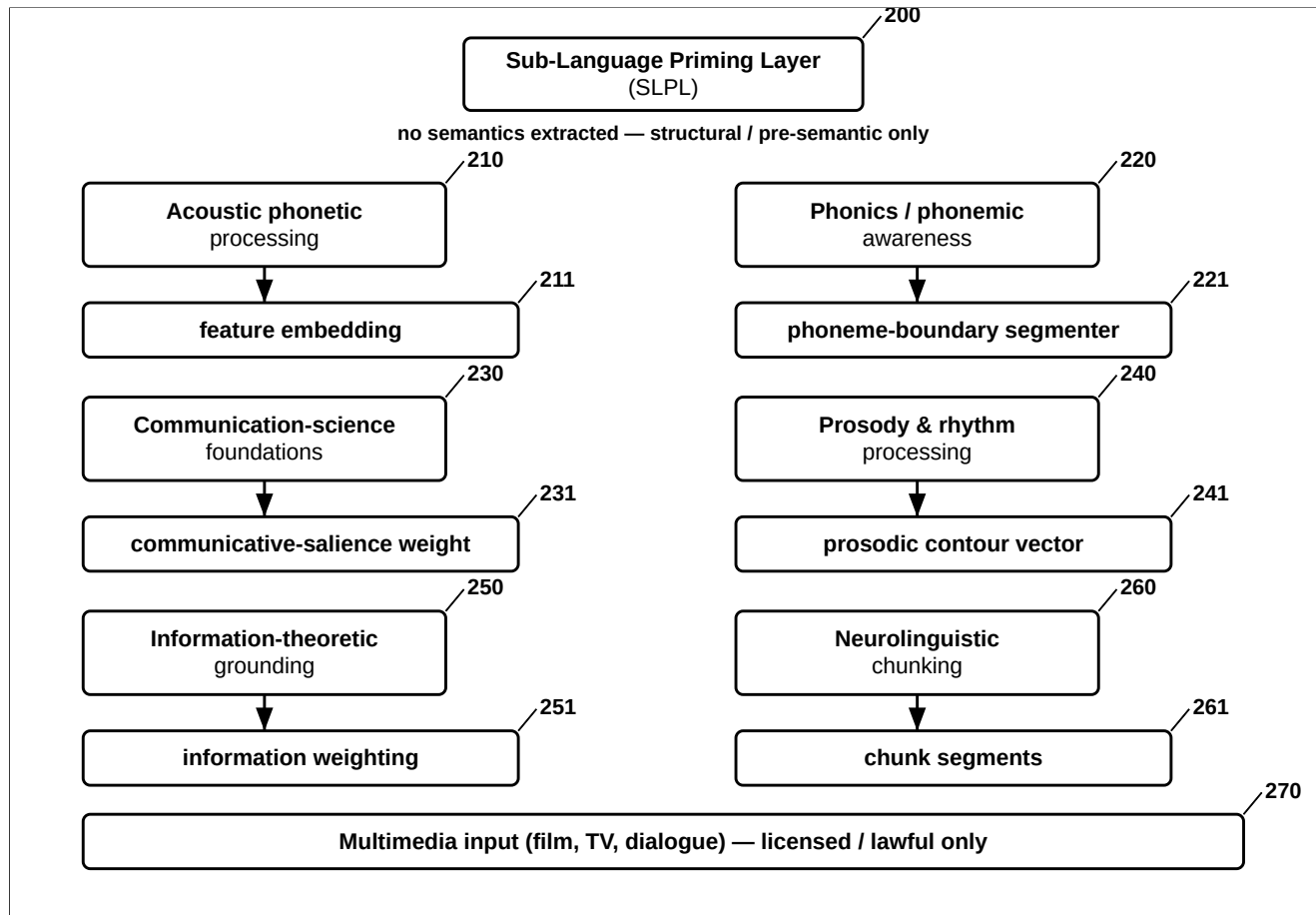


FIG. 2 — Sub-Language Priming Layer (SLPL) and Its Measurable Outputs

FIG. 3

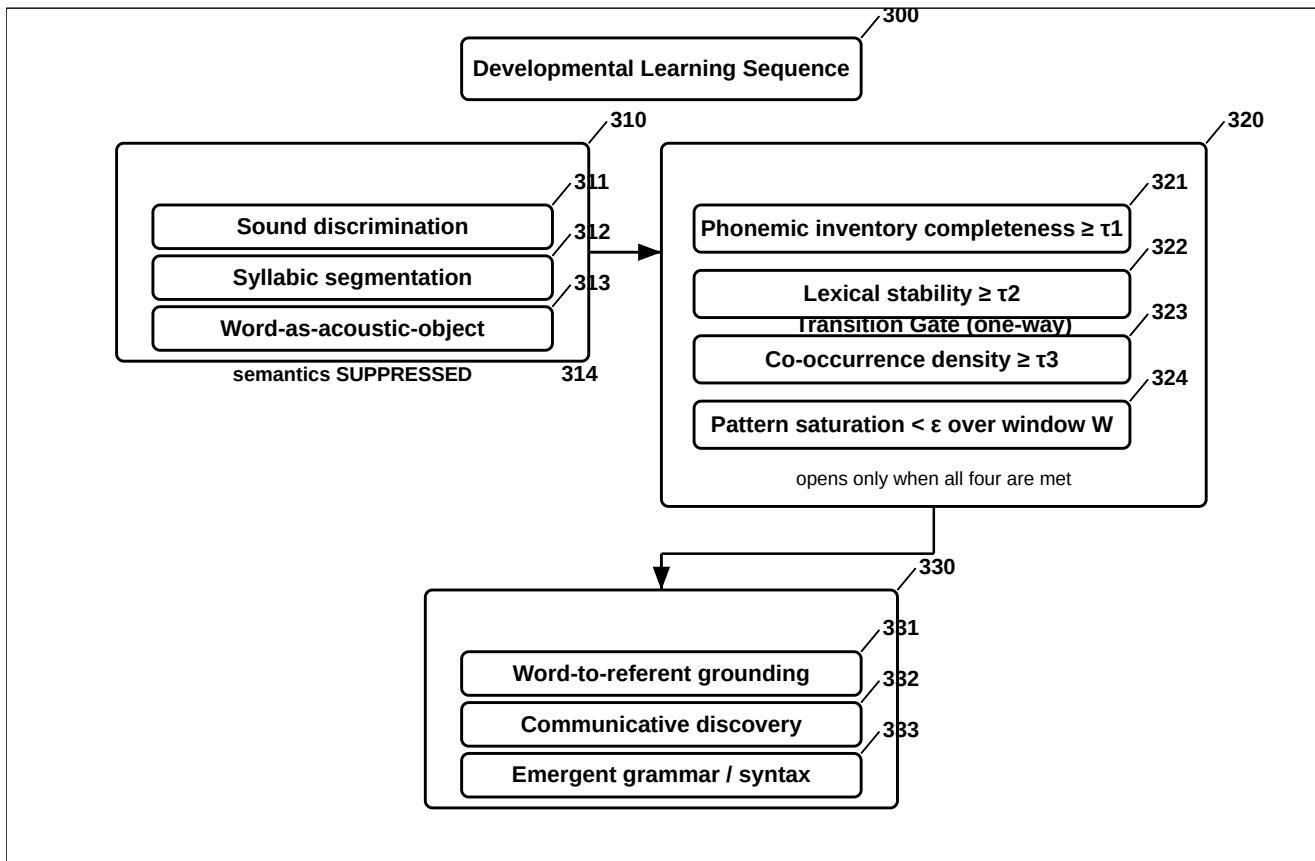


FIG. 3 — Developmental Learning Sequence and One-Way Transition Gate

FIG. 4

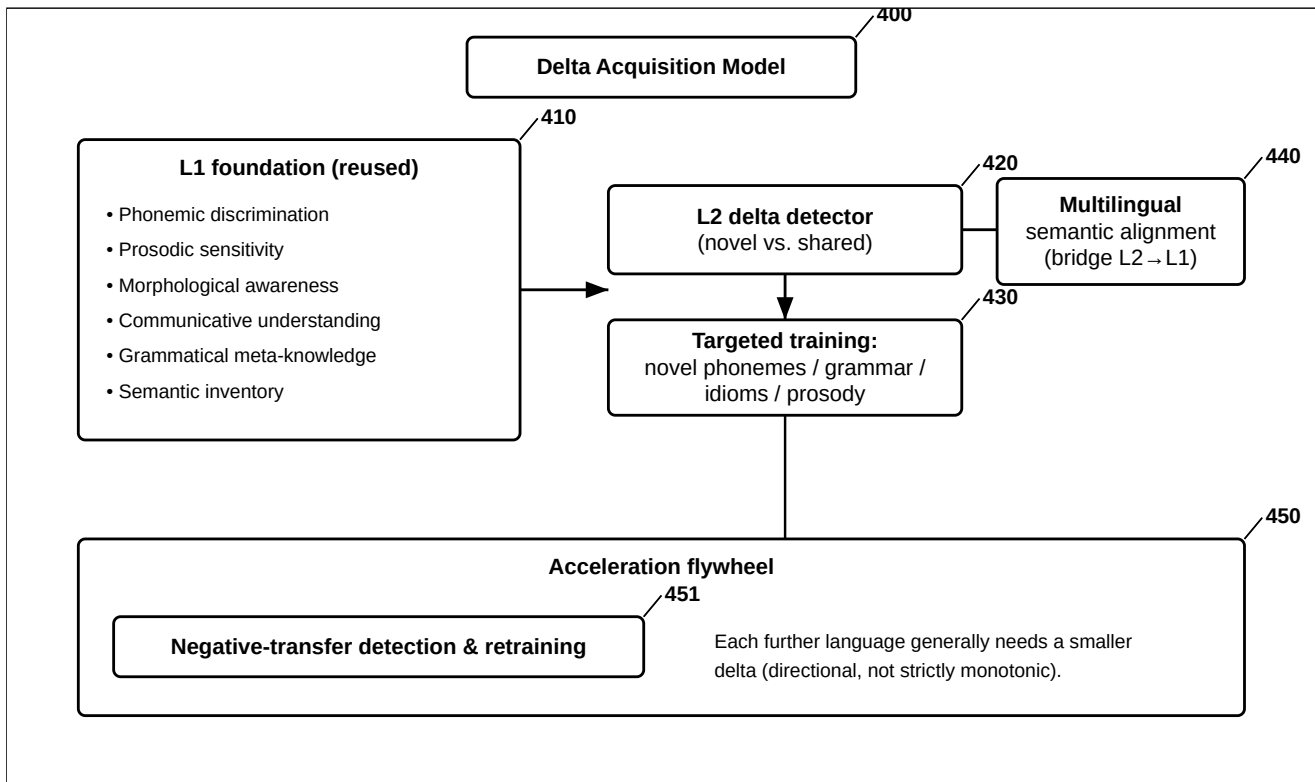


FIG. 4 — Multilingual Transfer — Delta Acquisition and Acceleration Flywheel

FIG. 5

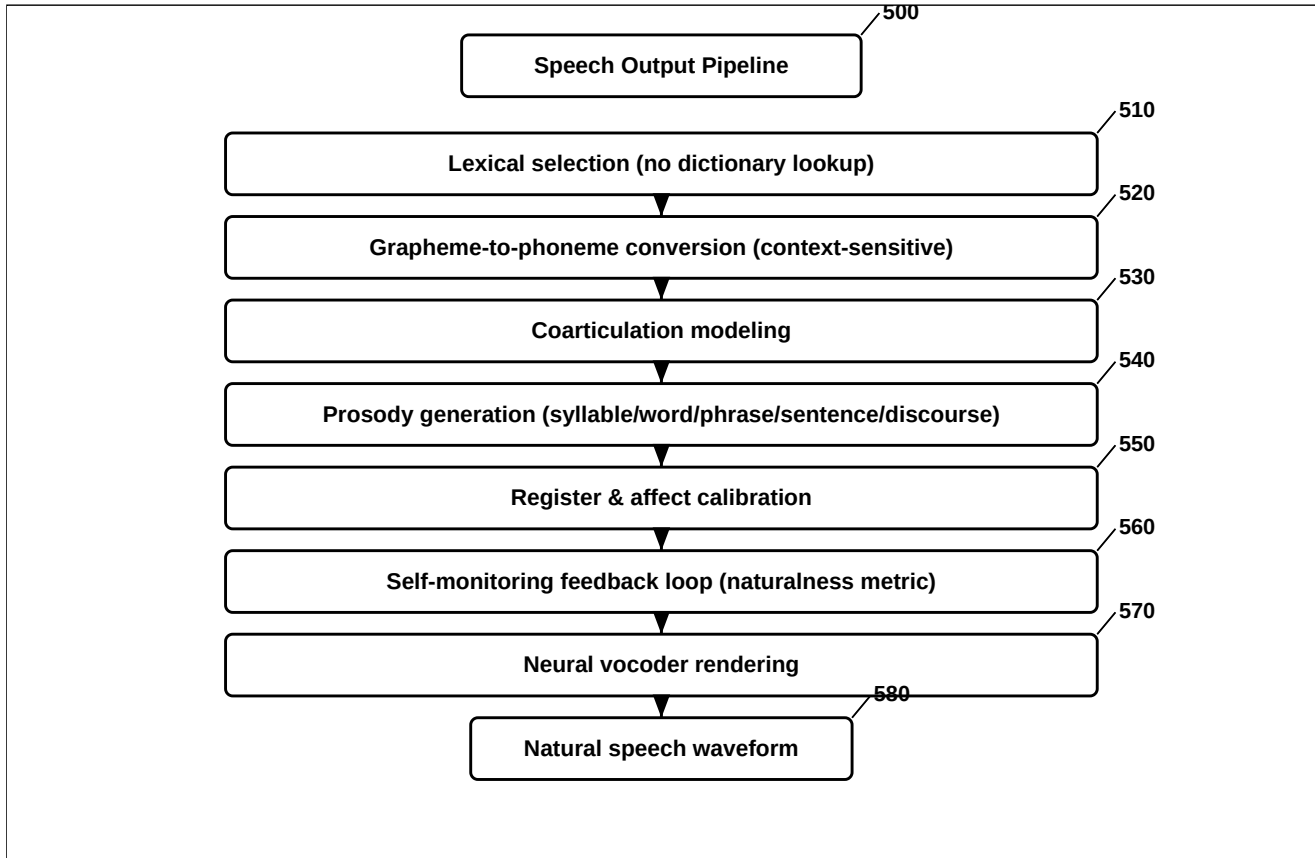


FIG. 5 — Speech Output Pipeline (Seven Learned Stages)

FIG. 6

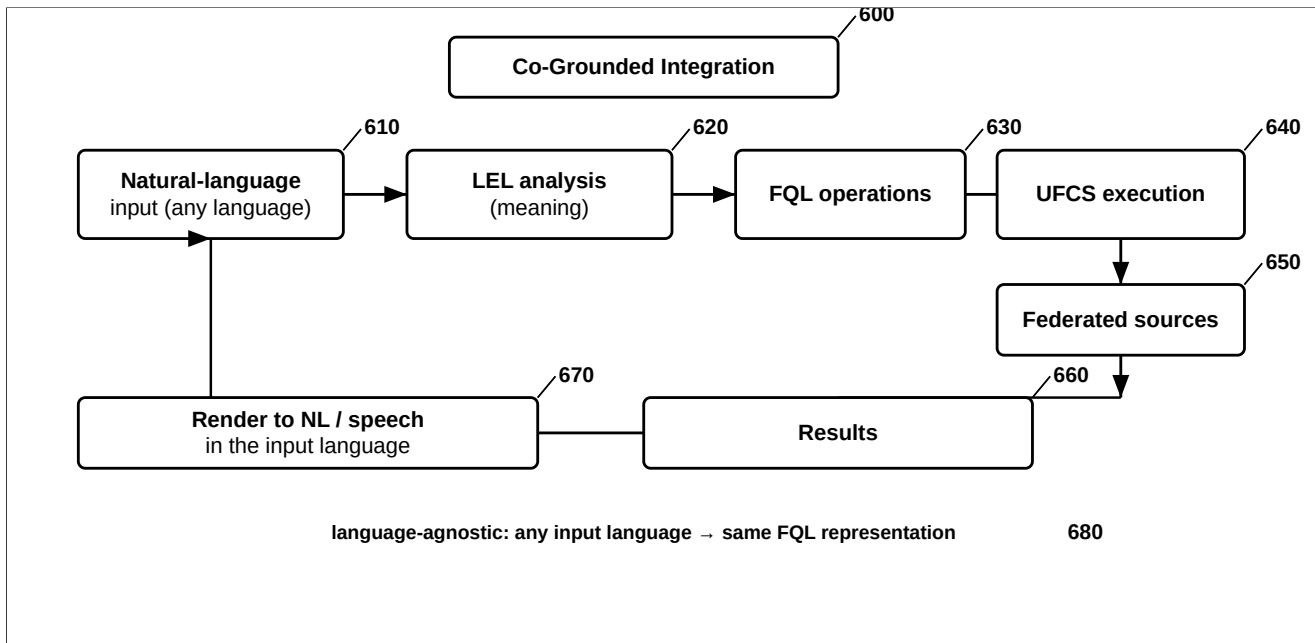


FIG. 6 — Co-Grounded Integration with FQL and UFCS

FIG. 7

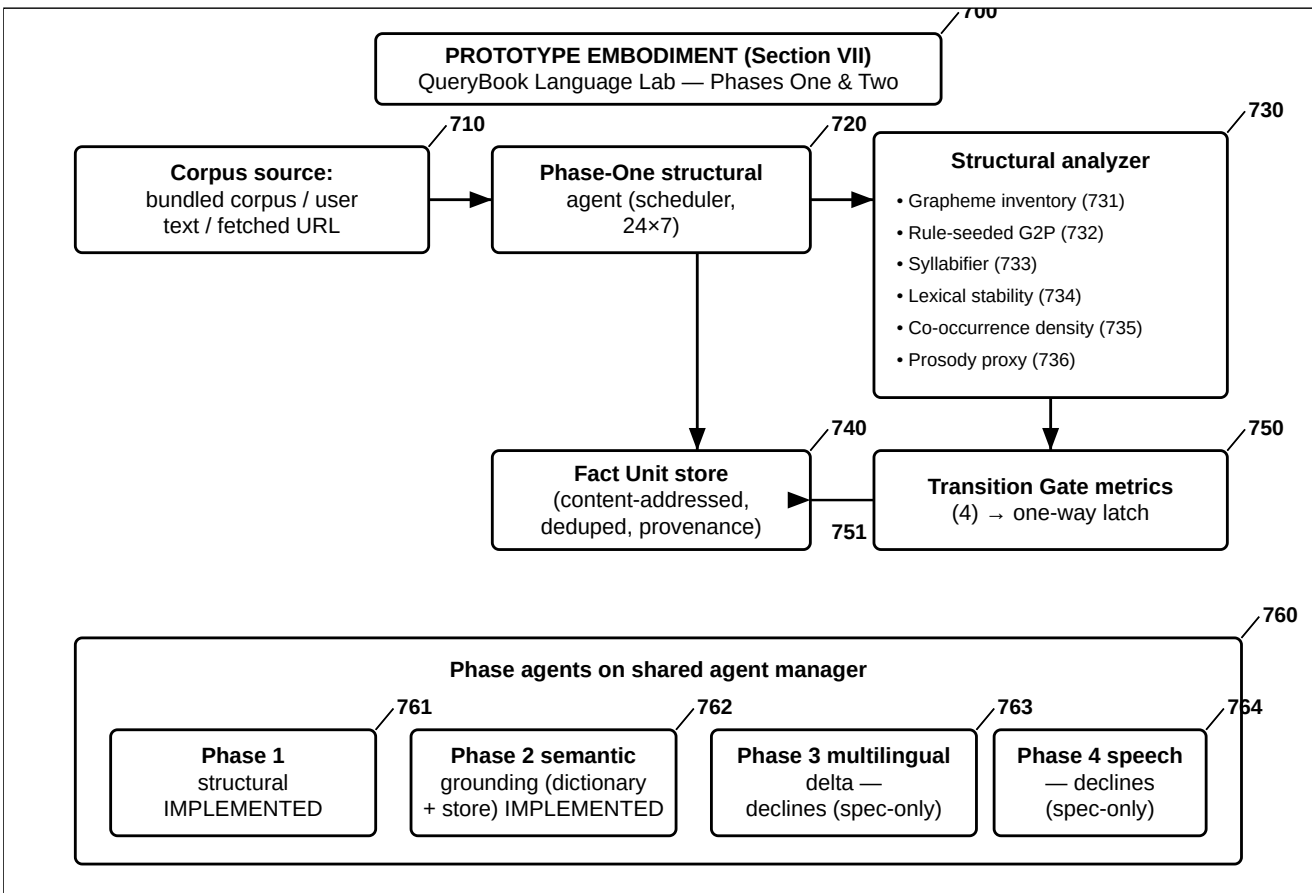


FIG. 7 — Reduced-to-Practice Embodiment — Language Lab Phases One and Two Prototype

FIG. 8

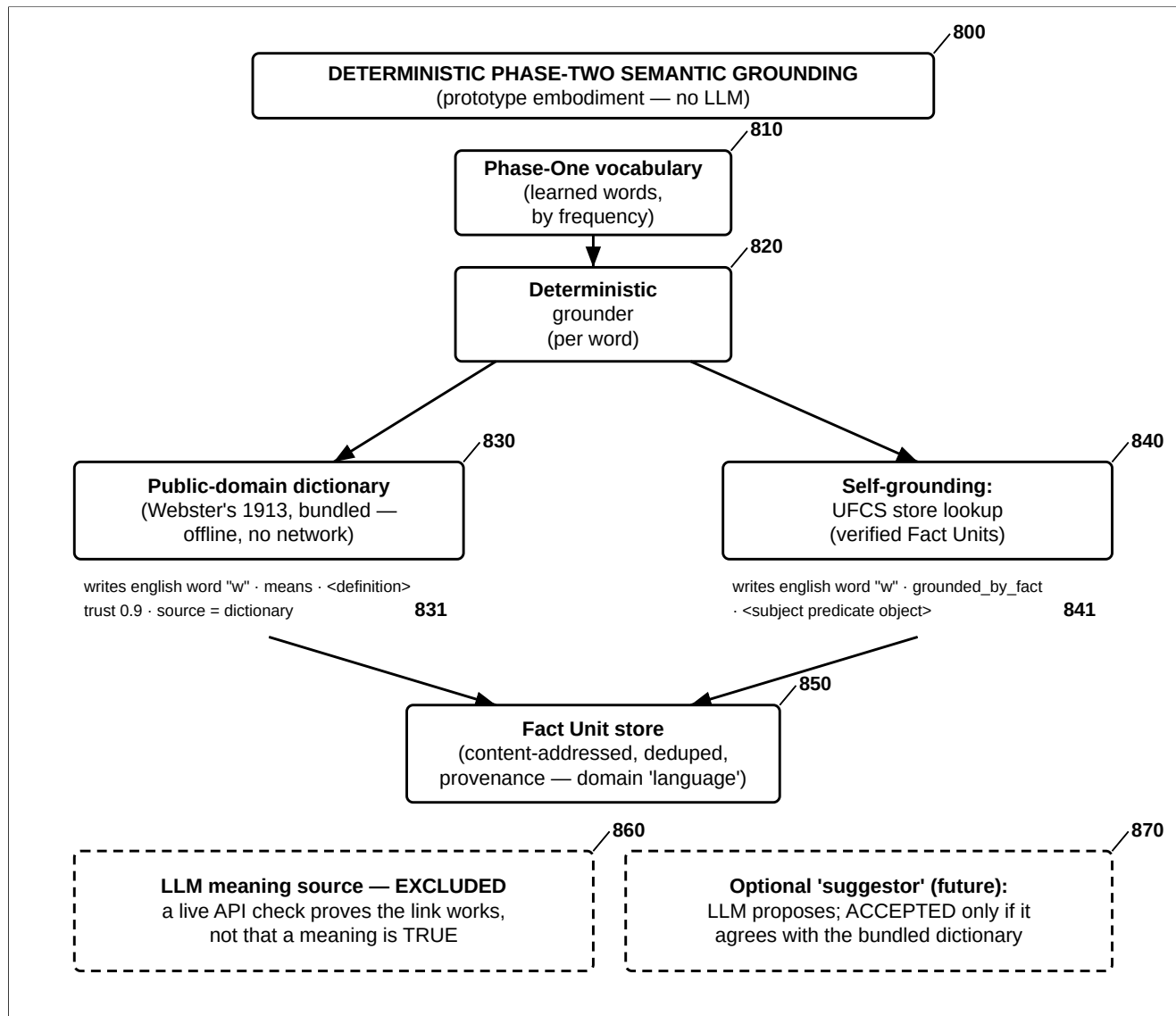


FIG. 8 — Deterministic Phase-Two Semantic Grounding — Dictionary and Store (No LLM)

## QueryBook Platform Drawings (FIG. 9–FIG. 30) — the system in which the language subsystem is integral

FIG. 9 through FIG. 30 depict the QueryBook deterministic, provenance-tracked federated knowledge and query platform of which the developmental language and speech subsystem (the Language Expression Layer, FIG. 1–FIG. 8 and FIG. 31–FIG. 32) is an integral, co-grounded part (see FIG. 6). Together with the language-subsystem figures these form **one QueryBook figure set**. Draft — the language-subsystem figures use a 100–999 numeral series and the platform figures a 100–2200 series; counsel/draftsperson to consolidate the two into a single QueryBook numeral series and finalize to 37 CFR 1.84 before filing.

FIG. 9

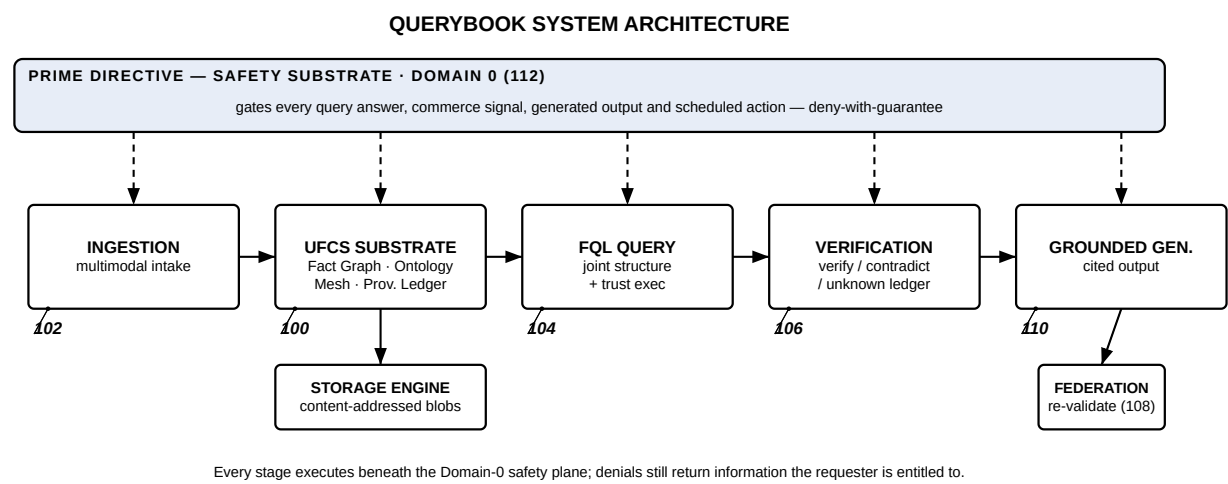


FIG. 9

FIG. 9 — QueryBook system architecture (Prime Directive as Domain 0)

FIG. 10

FACT UNIT — ATOM, ENVELOPE, AND CONTENT-DERIVED IDENTITY

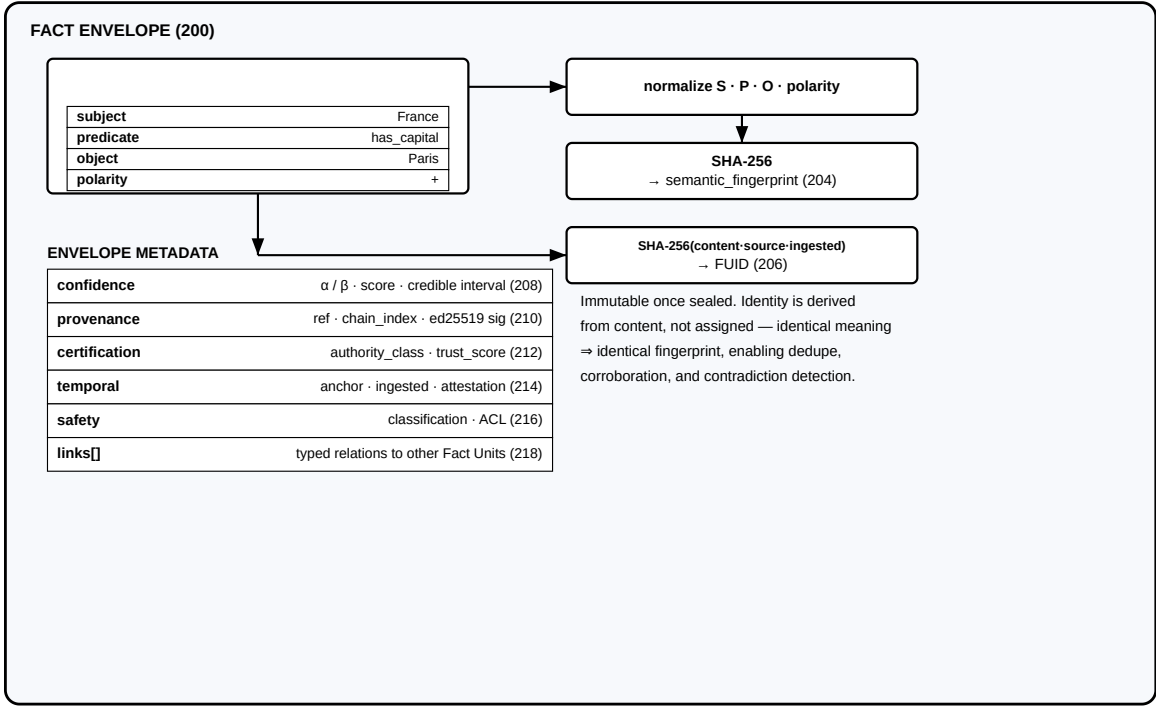


FIG. 10

FIG. 10 — Fact Unit — atom, envelope, and content-derived identity

FIG. 11

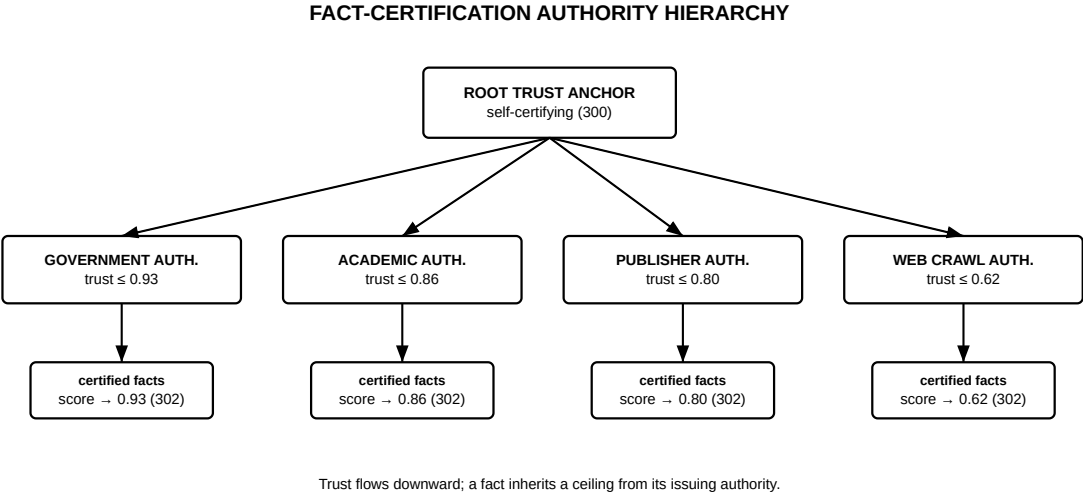


FIG. 11

FIG. 11 — Fact-certification authority hierarchy

FIG. 12

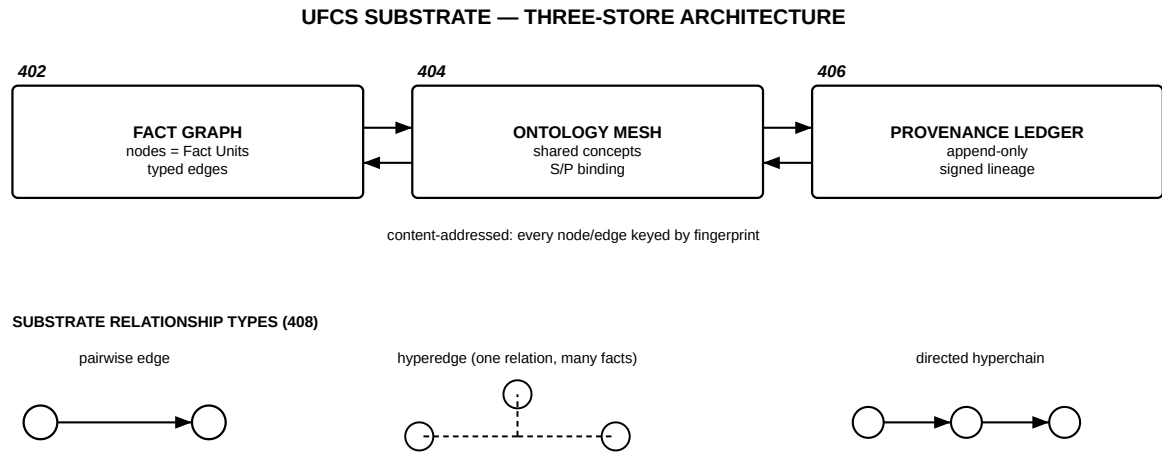
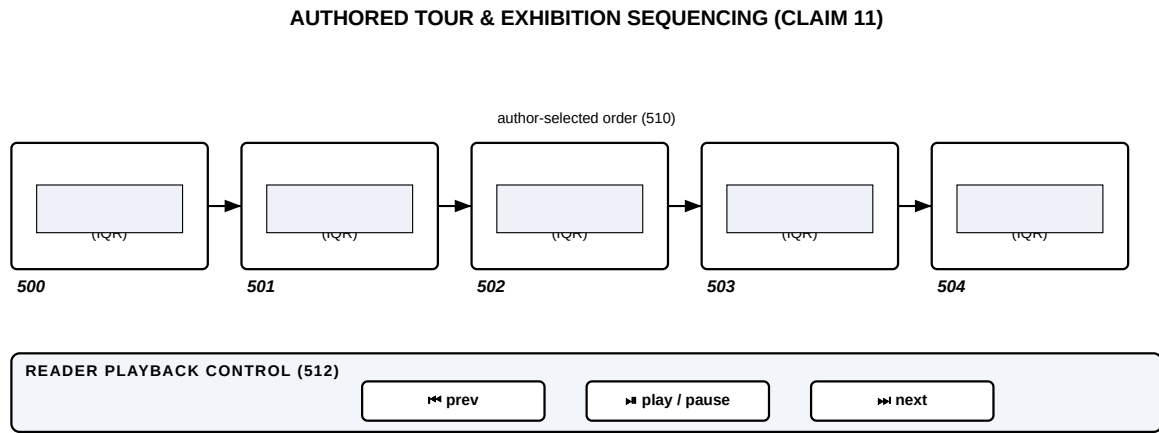


FIG. 12

FIG. 12 — UFCS substrate — three-store architecture

FIG. 13



Author fixes the sequence; the reader controls pace and direction along the ordered waypoints.

FIG. 13

FIG. 13 — Authored tour and exhibition sequencing

FIG. 14

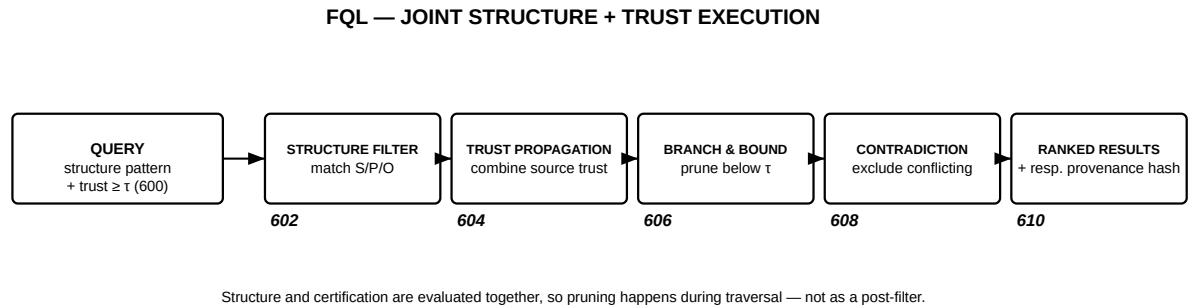


FIG. 14

FIG. 14 — FQL — joint structure and trust execution

FIG. 15

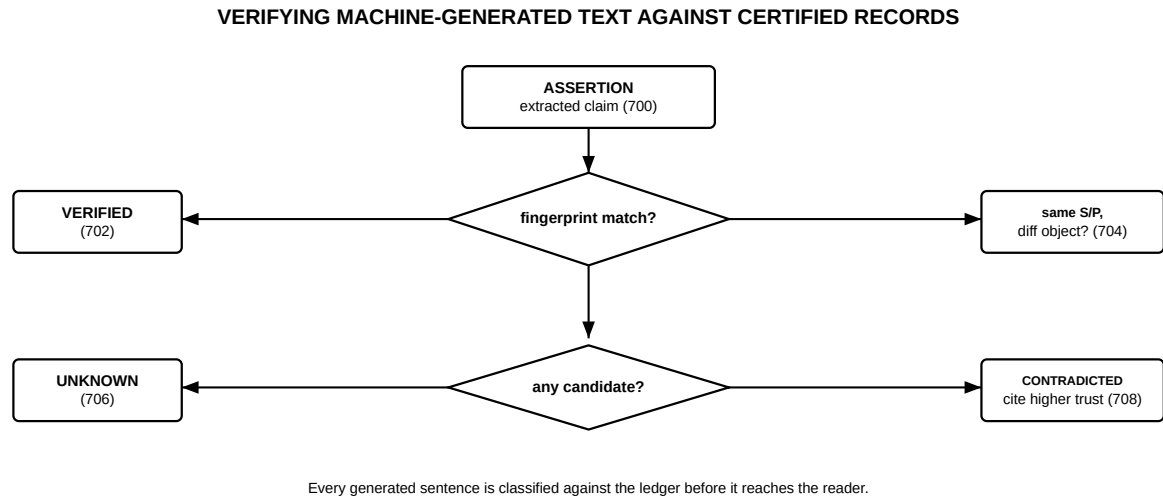


FIG. 15

FIG. 15 — Verifying machine-generated text against certified records

FIG. 16

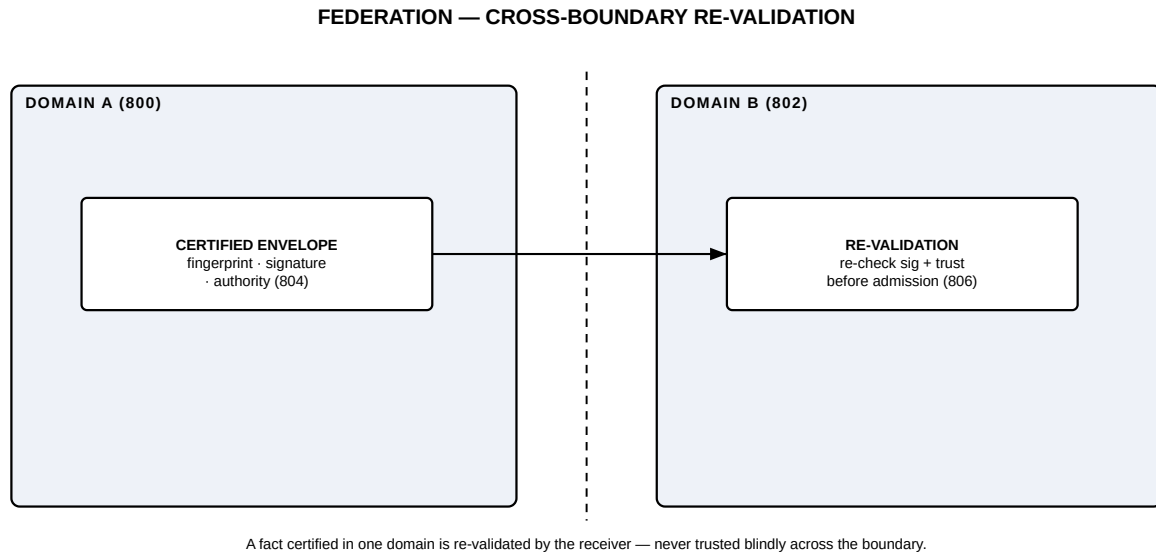
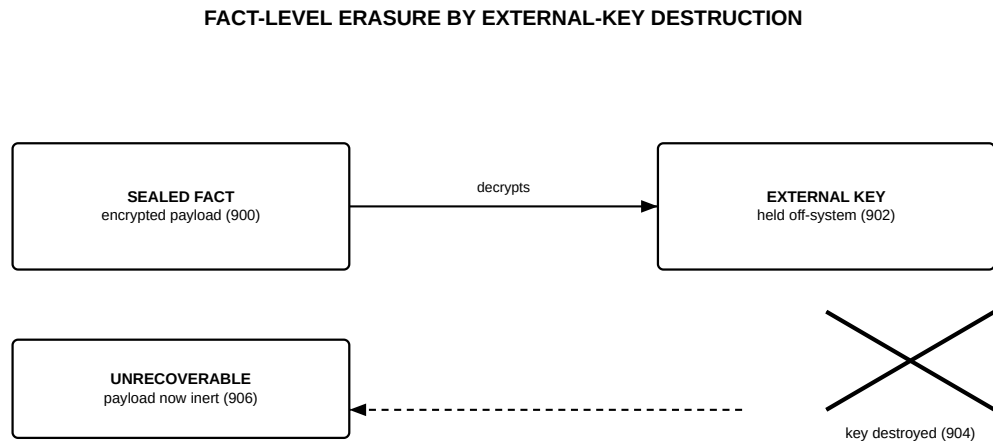


FIG. 16

FIG. 16 — Federation — cross-boundary re-validation

FIG. 17



Erase one fact by destroying its key; the content-addressed store is otherwise append-only and immutable.

FIG. 17

FIG. 17 — Fact-level erasure by external-key destruction

FIG. 18

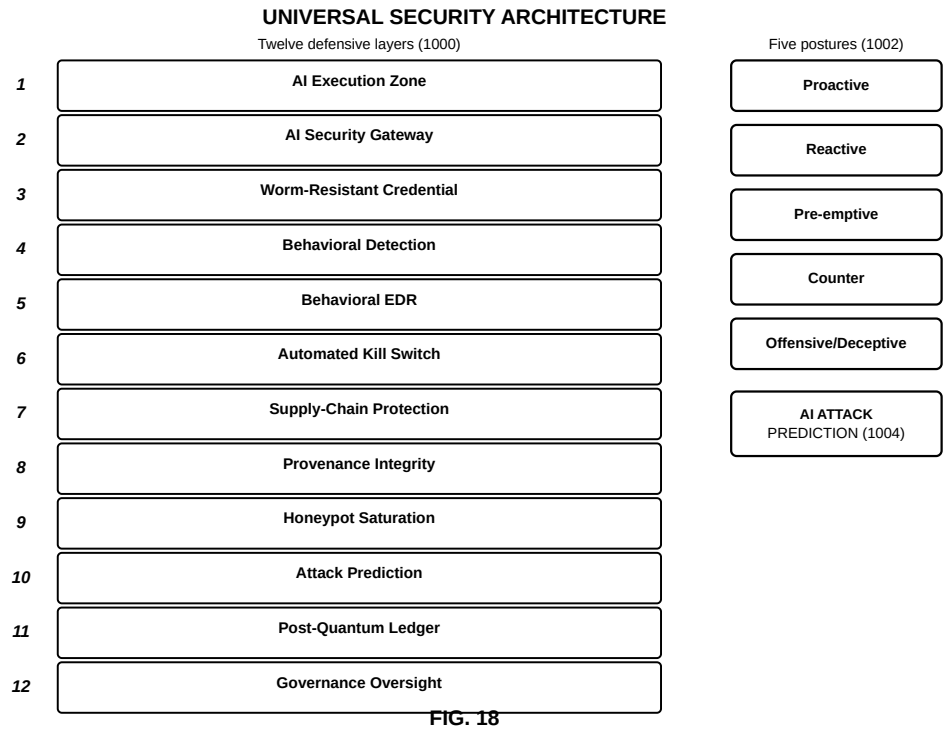


FIG. 18 — Universal security architecture

FIG. 19

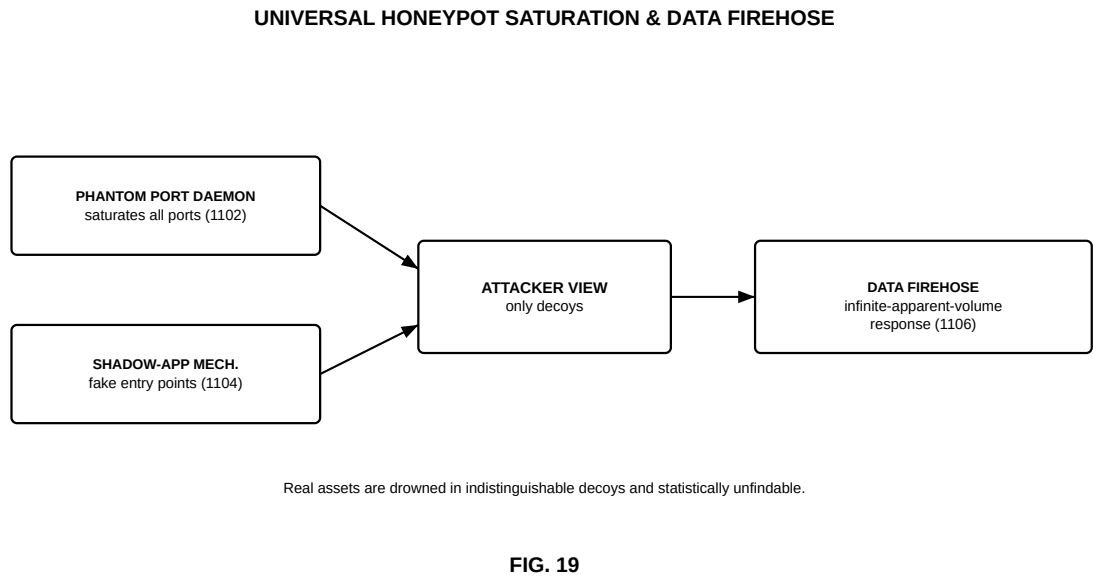


FIG. 19 — Universal honeypot saturation and data firehose

FIG. 20

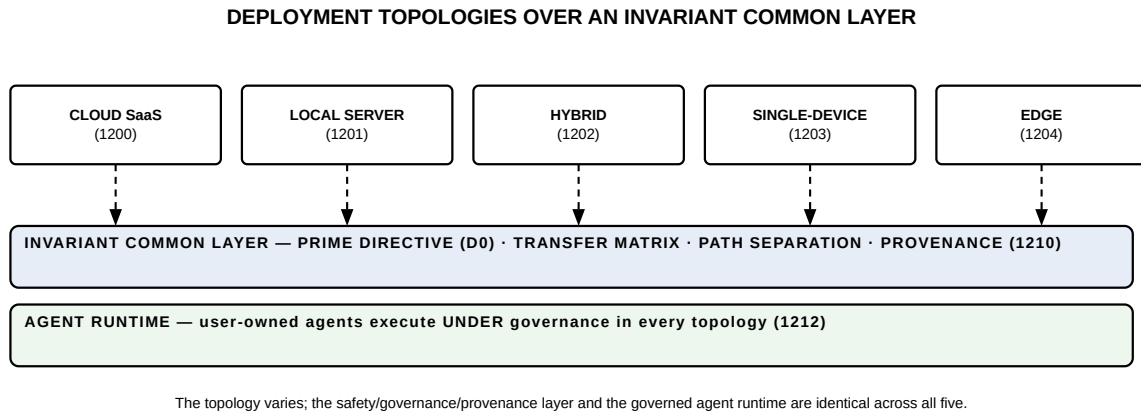


FIG. 20

FIG. 20 — Deployment topologies over an invariant common layer

FIG. 21

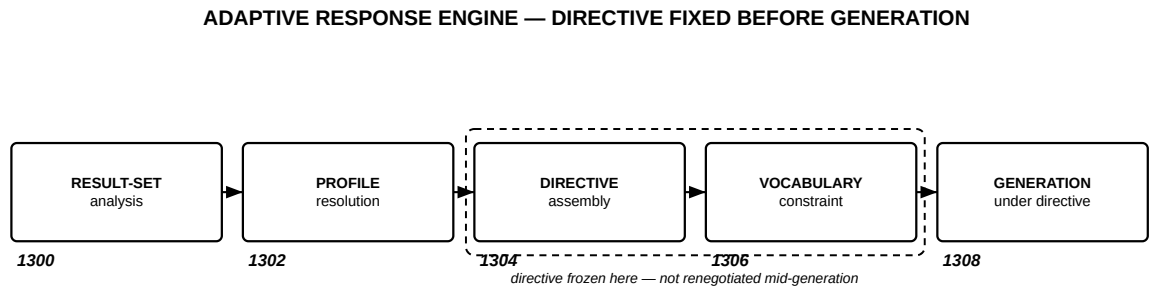
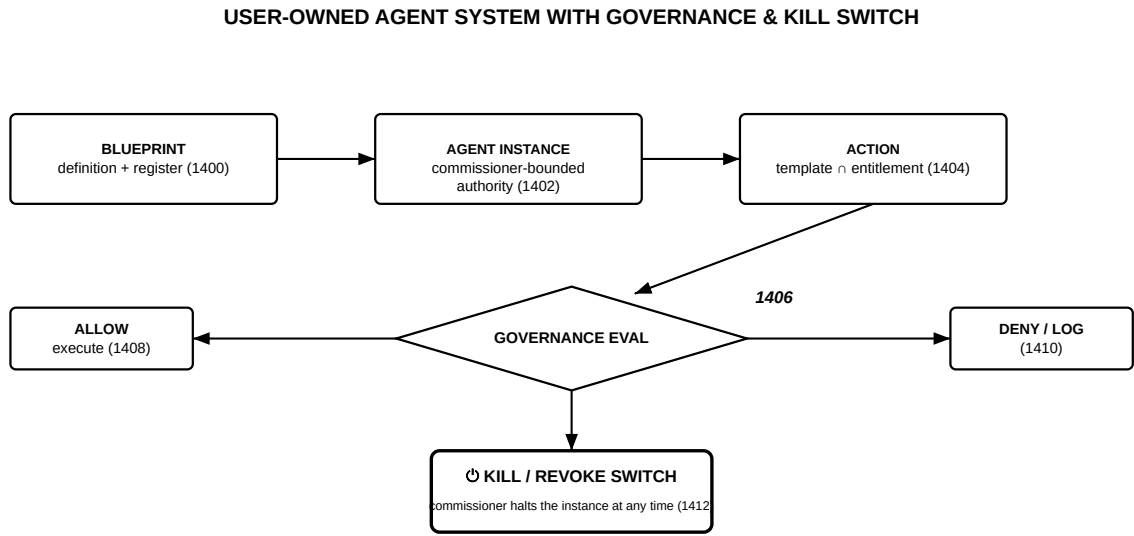


FIG. 21

FIG. 21 — Adaptive response engine — directive fixed before generation

FIG. 22

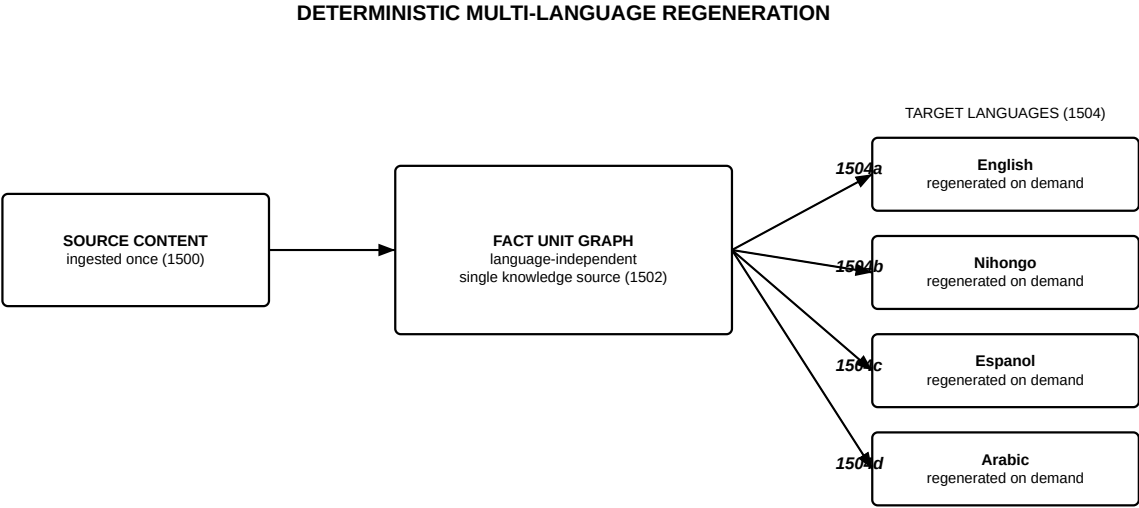


Governance is checked at every action; an agent can never exceed its commissioner, who holds an unconditional kill control.

FIG. 22

FIG. 22 — User-owned agent system with governance and kill switch

FIG. 23



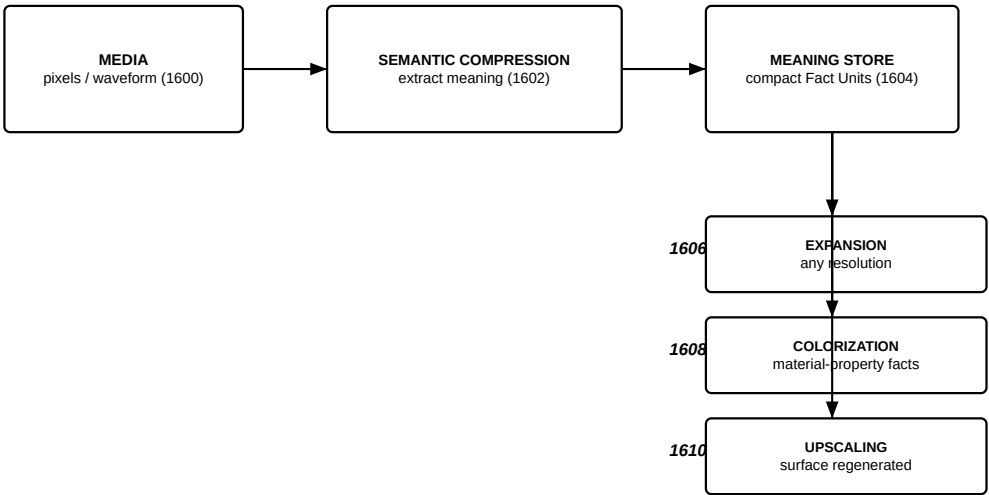
One graph, many surfaces: no per-language re-ingestion; each language is a projection of the same facts.

FIG. 23

FIG. 23 — Deterministic multi-language regeneration

FIG. 24

SEMANTIC COMPRESSION, EXPANSION, COLORIZATION & UPSCALING



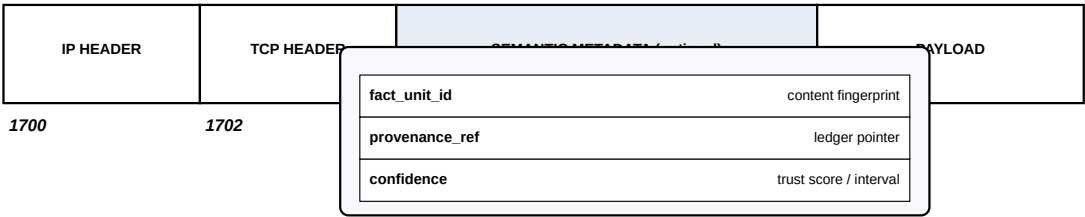
Store meaning, not surface; compression and regeneration are the same primitive — no generational quality loss.

FIG. 24

FIG. 24 — Semantic compression, expansion, colorization and upscaling

FIG. 25

QUERYBOOK-ENHANCED TCP/IP HYBRID PROTOCOL PACKET



A non-participating endpoint ignores the semantic header — provenance travels with the bits, backward-compatible.

FIG. 25

FIG. 25 — QueryBook-enhanced TCP/IP hybrid protocol packet

FIG. 26

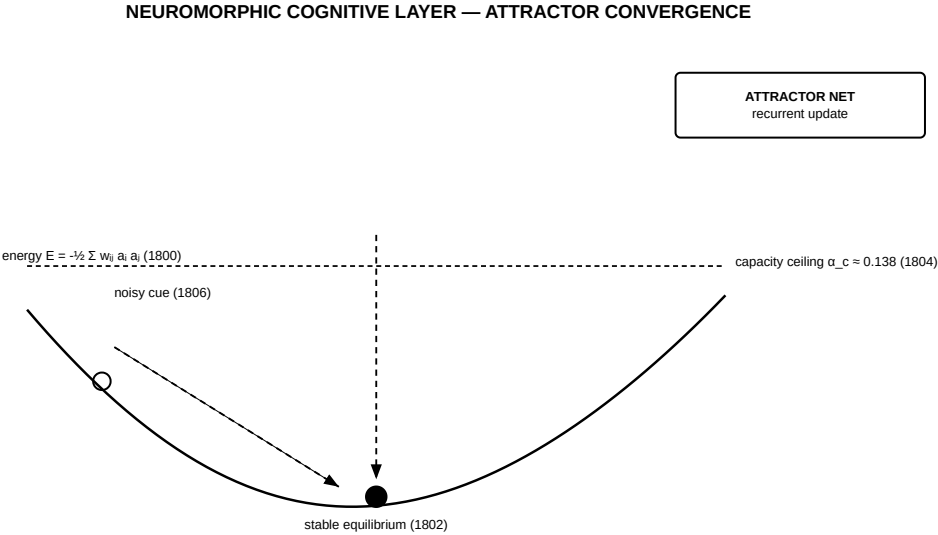
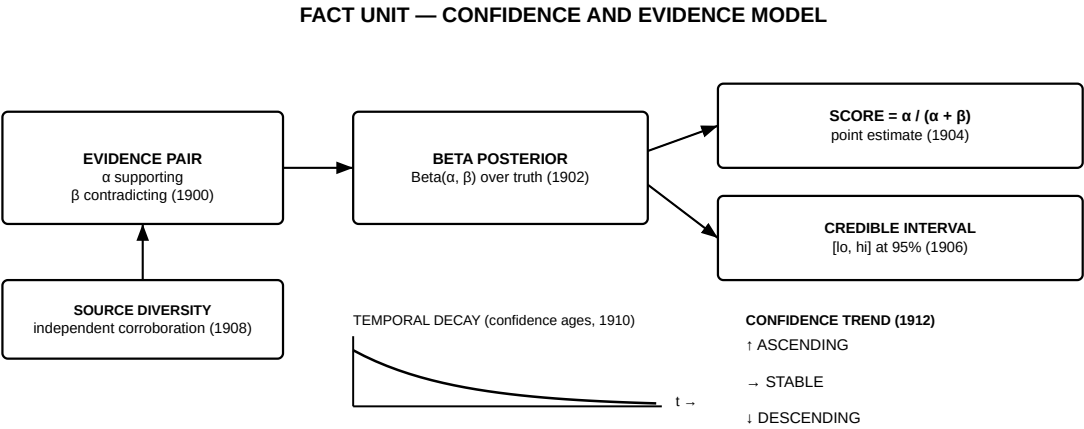


FIG. 26

A partial/noisy cue relaxes down the energy surface to the nearest stored equilibrium — recall as convergence, bounded by  $\alpha_c$ .

FIG. 26 — Neuromorphic cognitive layer — attractor convergence

FIG. 27



Trust is a governed evidence estimate — not a single number: it carries an interval, ages over time, and reports its direction.

FIG. 27

FIG. 27 — Fact Unit — confidence and evidence model

FIG. 28

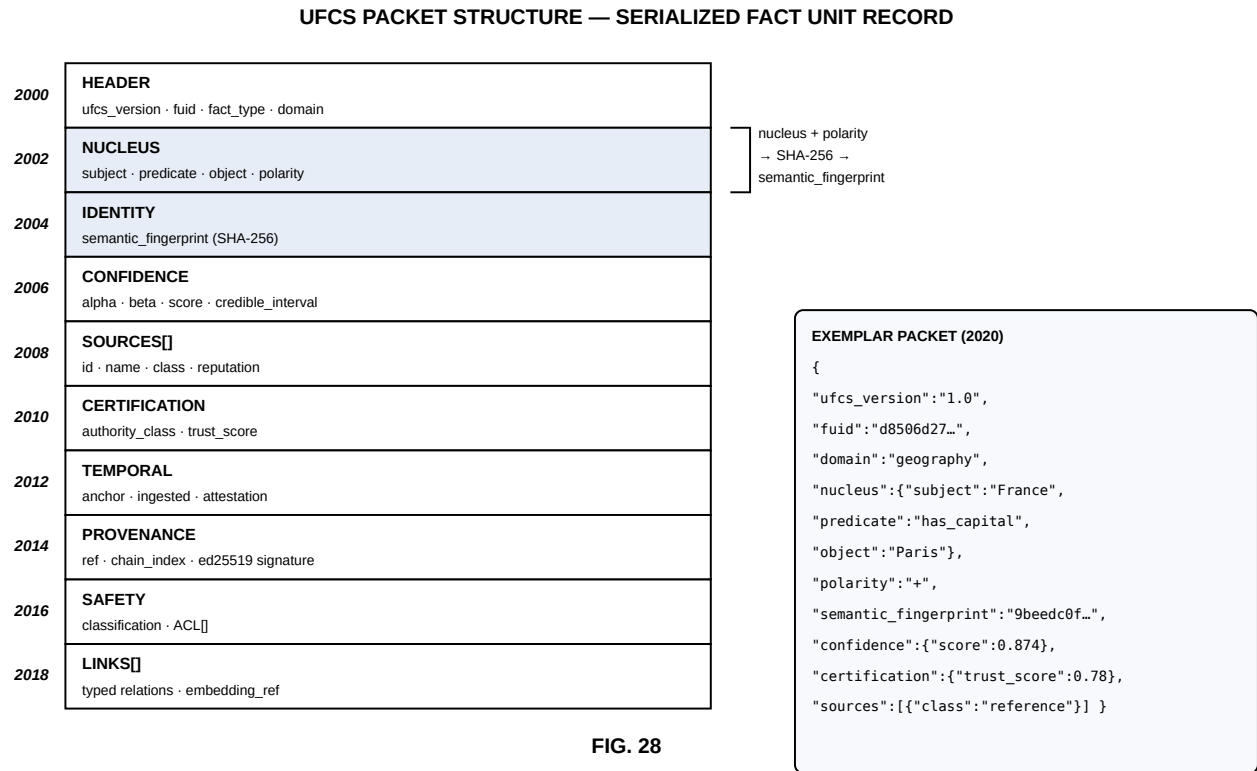


FIG. 28 — UFCS packet structure — serialized Fact Unit record

FIG. 29

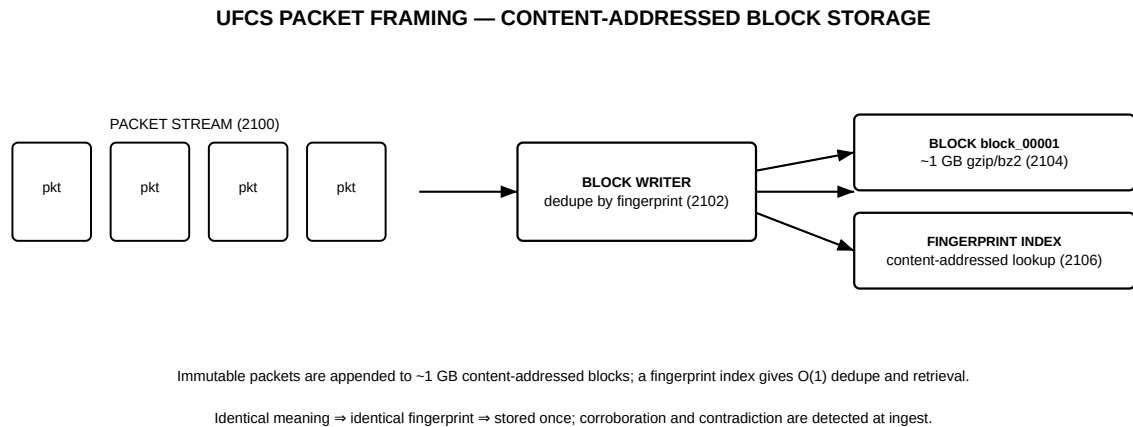


FIG. 29 — UFCS packet framing — content-addressed block storage

FIG. 30

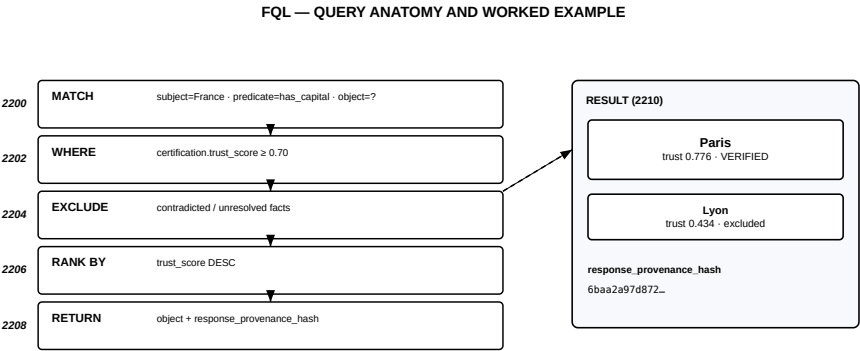


FIG. 30

Structure and trust are one query; the engine returns the highest-trust answer with a hash binding the result to its evidence.

FIG. 30 — FQL — query anatomy and worked example

FIG. 31

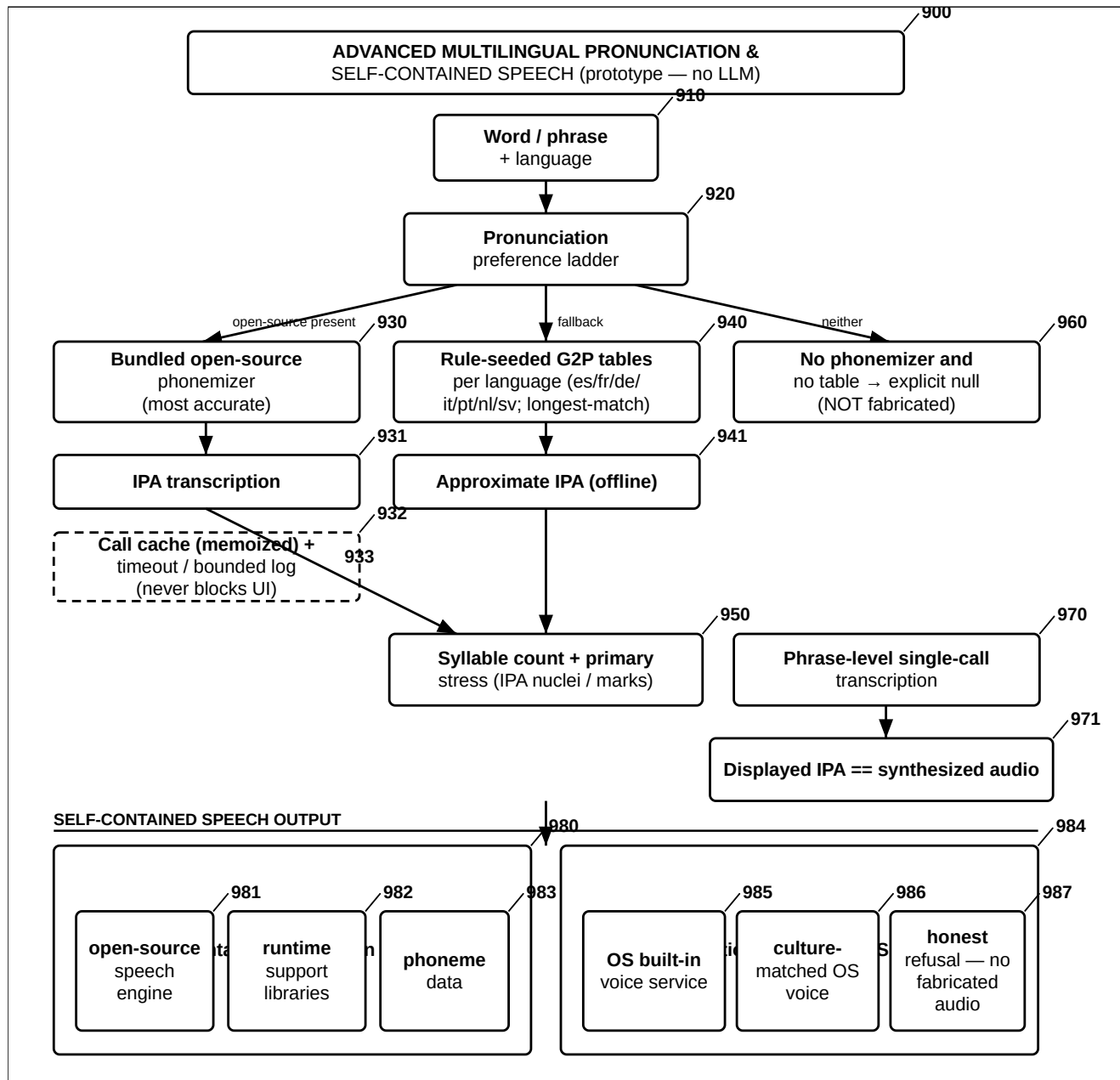


FIG. 31 — Advanced Multilingual Pronunciation and Self-Contained Speech (No LLM)

FIG. 32

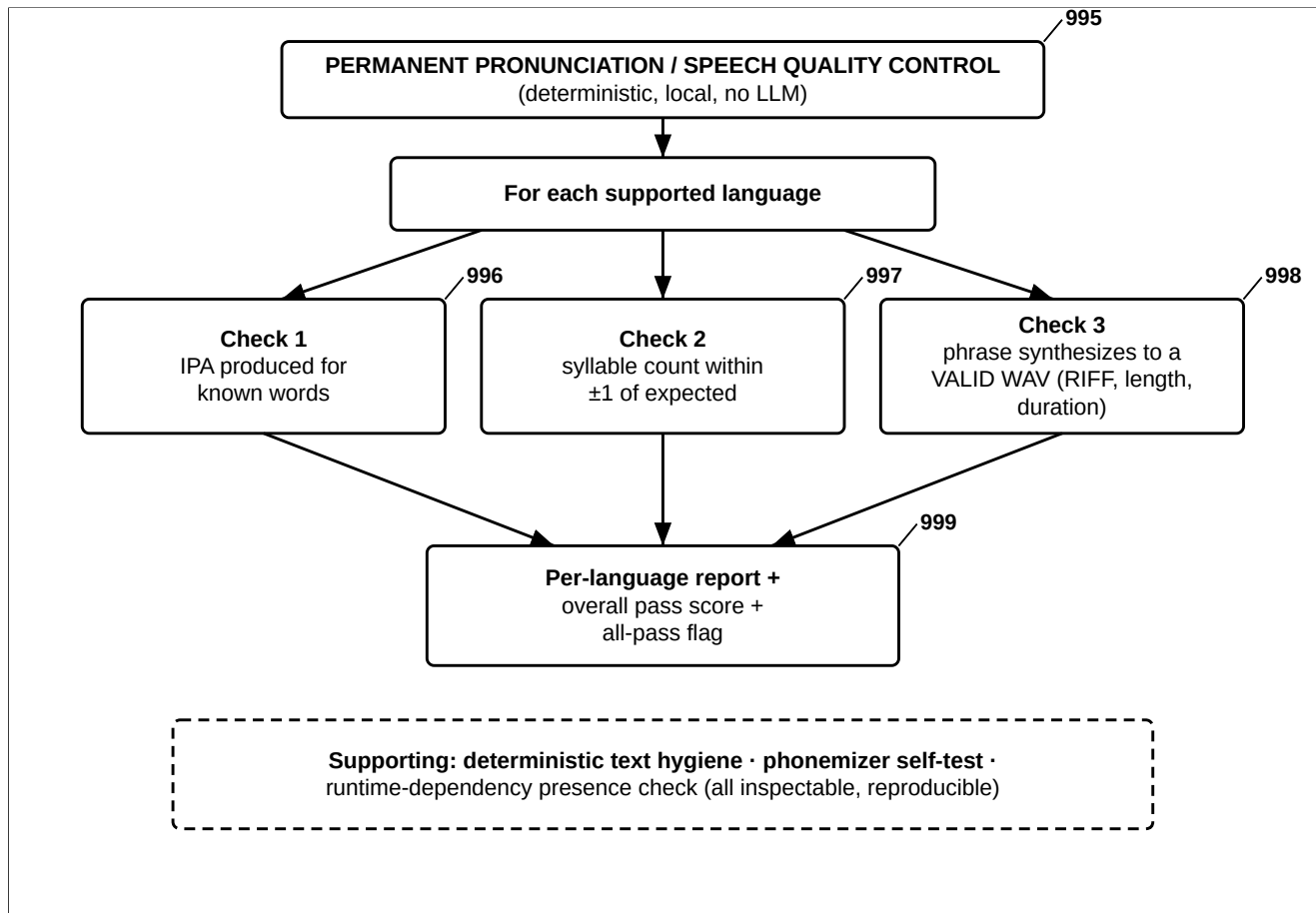


FIG. 32 — Permanent Pronunciation and Speech Quality-Control Battery (No LLM)

FIG. 33

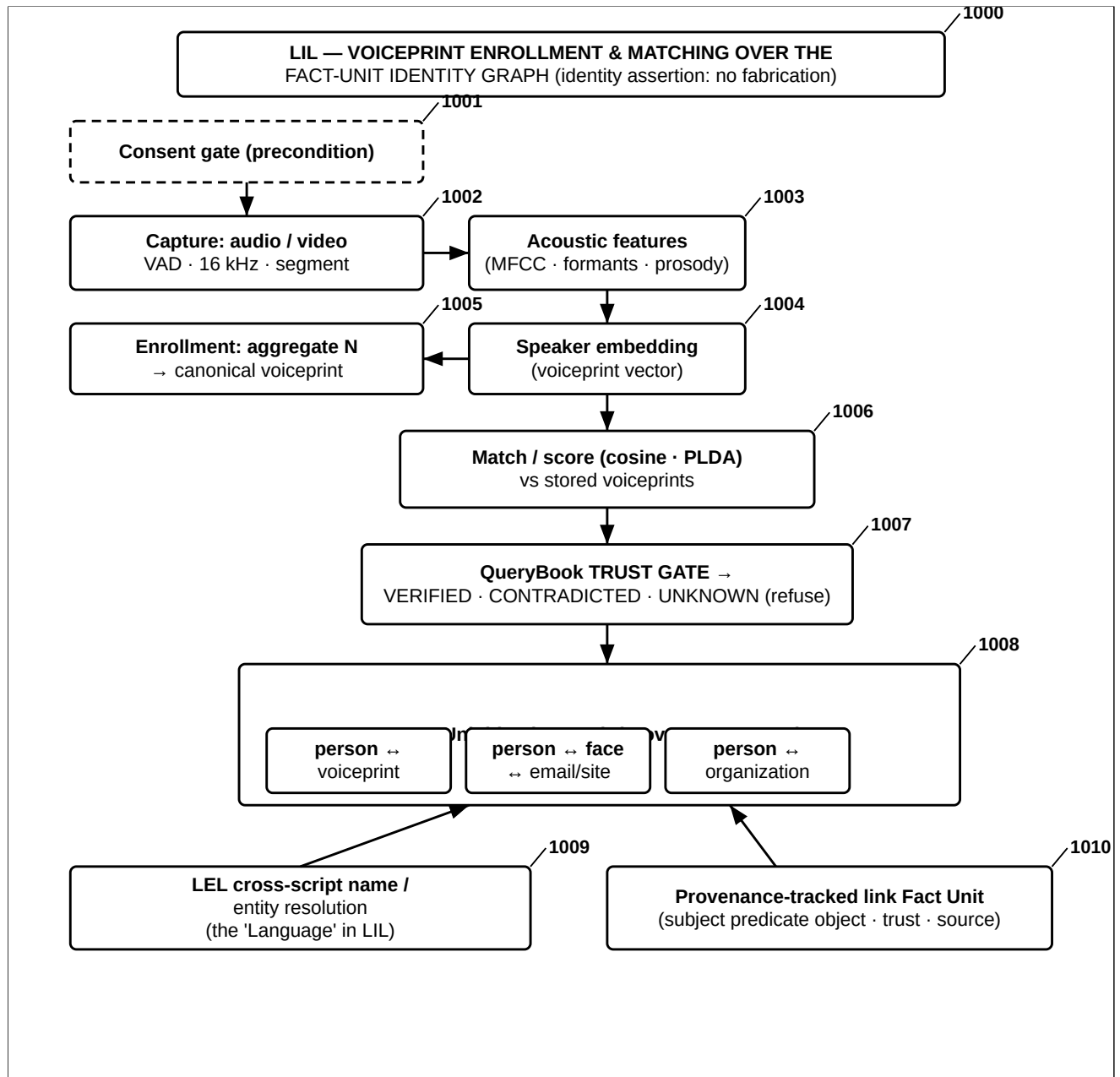


FIG. 33 — LIL — Voiceprint Enrollment and Matching over the Fact-Unit Identity Graph

FIG. 34

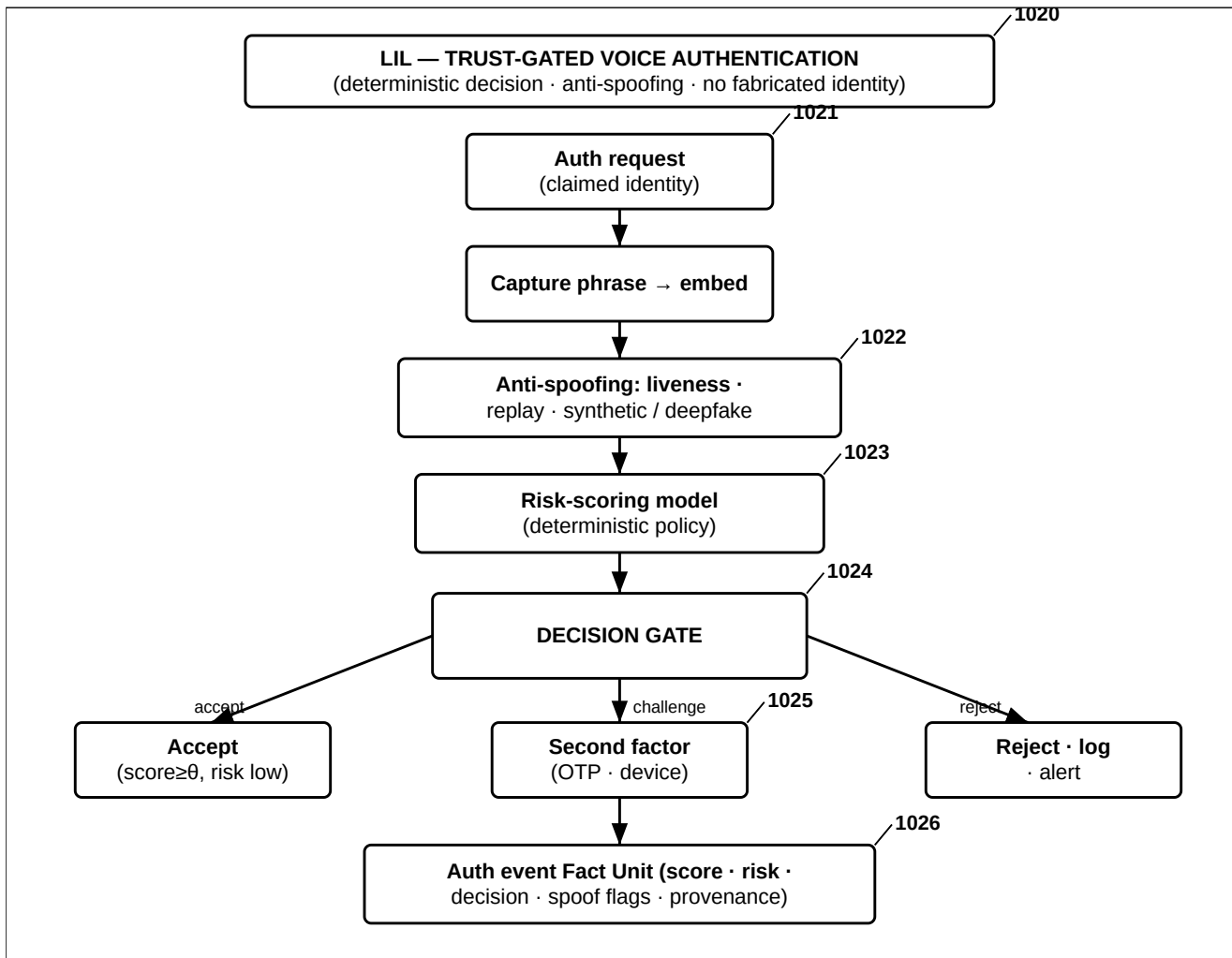


FIG. 34 — LIL — Trust-Gated Voice Authentication with Anti-Spoofing

FIG. 35

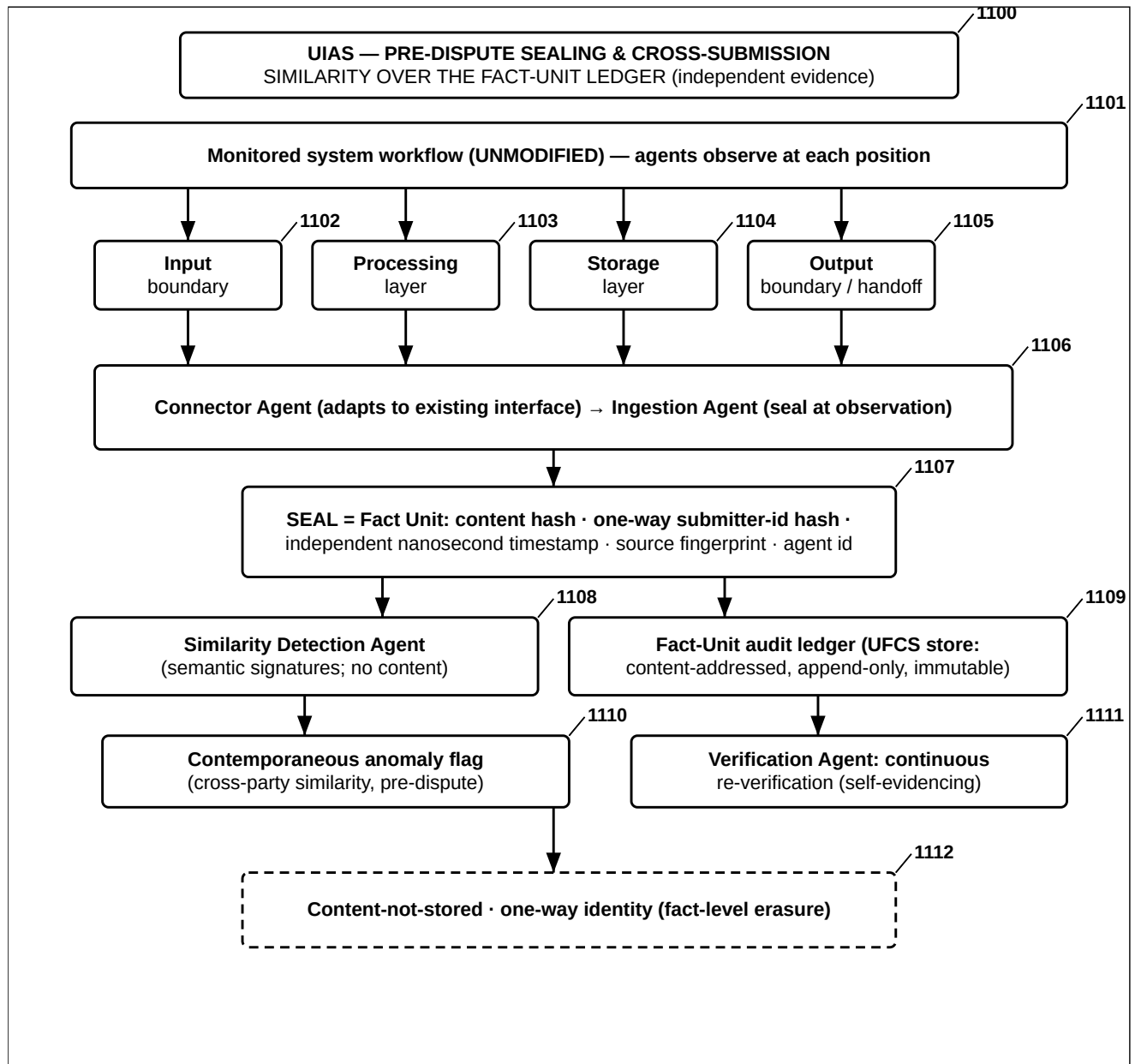


FIG. 35 — UIAS — Pre-Dispute Sealing and Cross-Submission Similarity over the Fact-Unit Ledger

FIG. 36

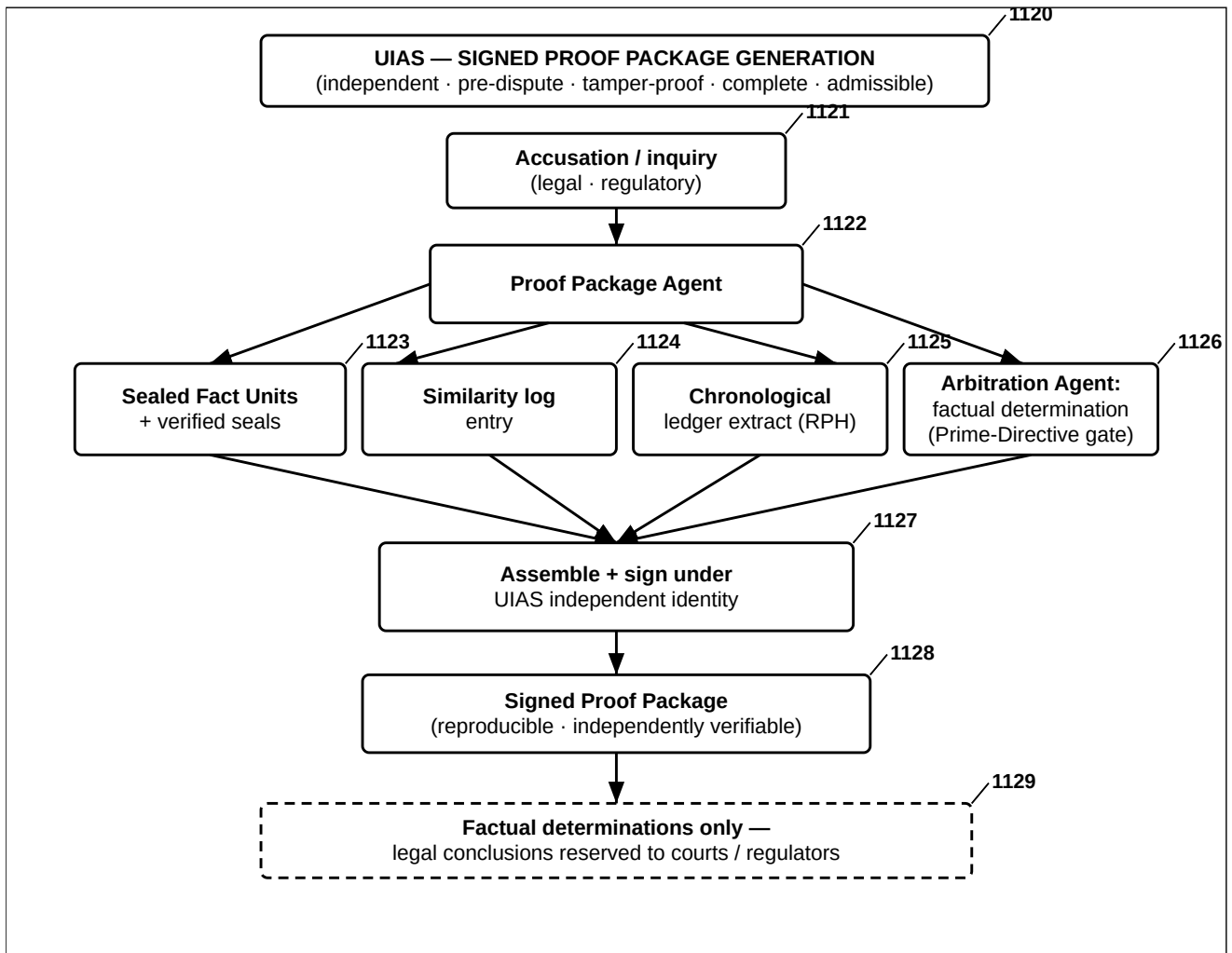


FIG. 36 — UIAS — Signed Proof Package Generation

FIG. 37

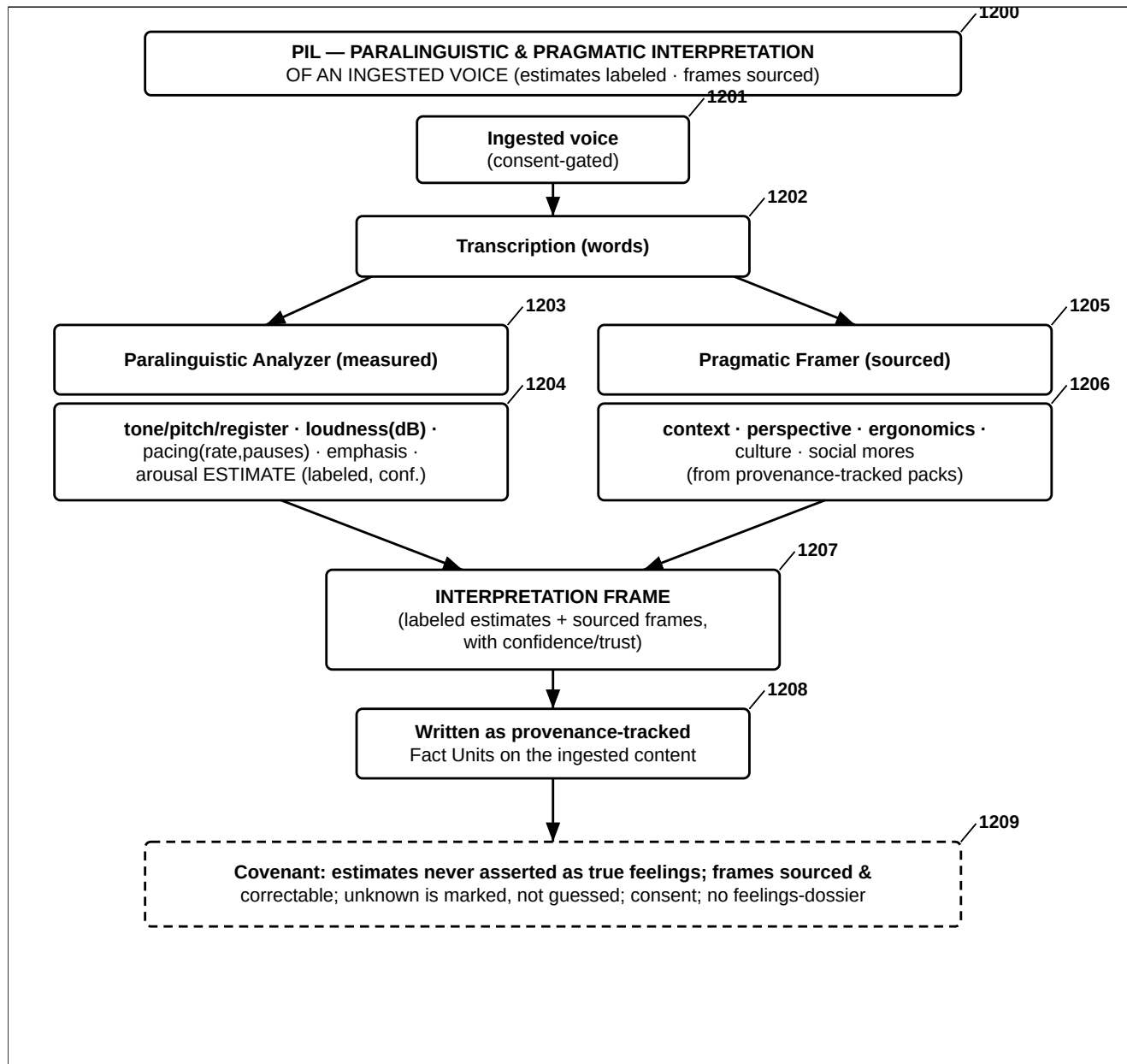


FIG. 37 — PIL — Paralinguistic and Pragmatic Interpretation of an Ingested Voice

FIG. 38

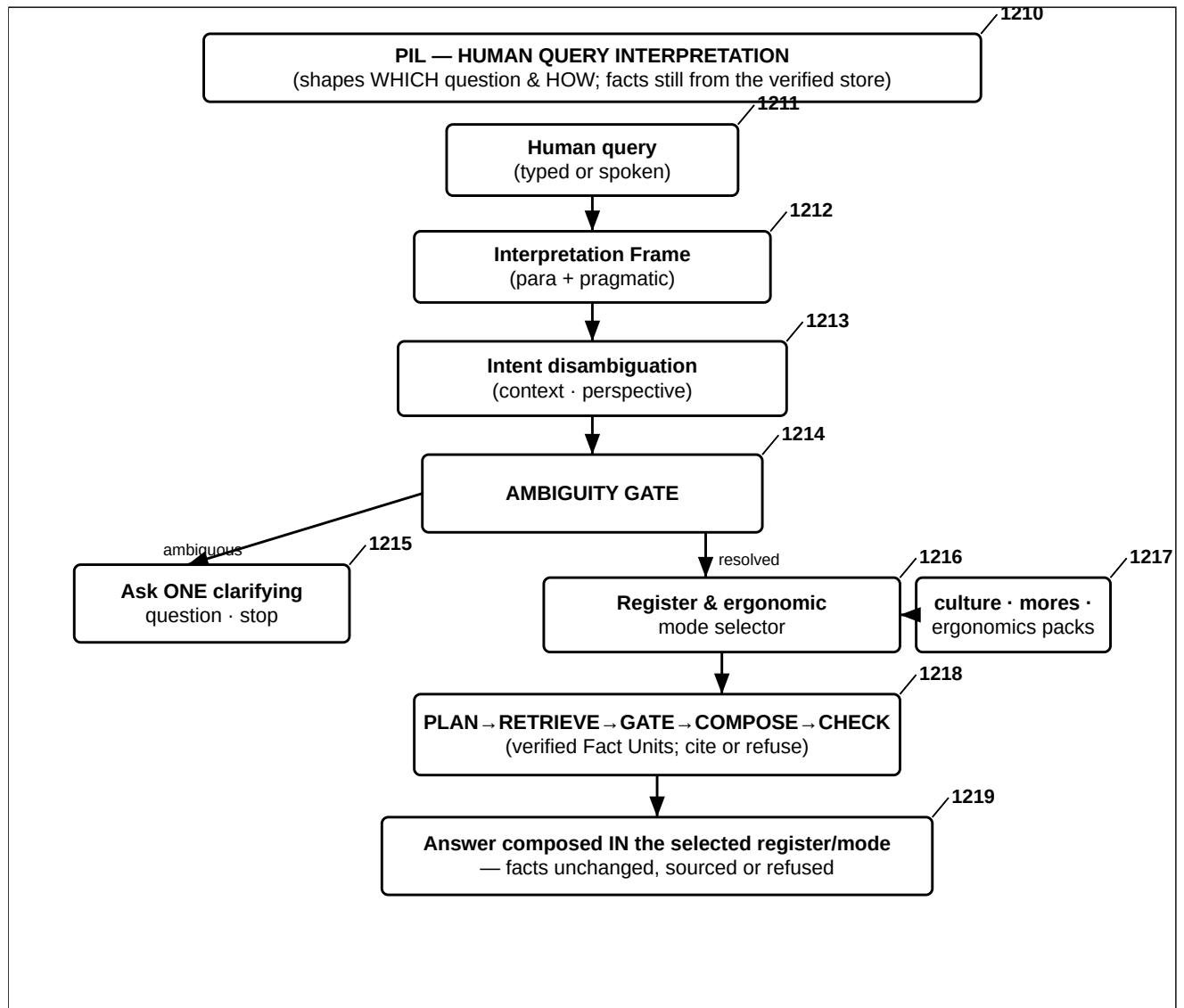


FIG. 38 — PIL — Human Query Interpretation and Covenant-Bound Answering

FIG. 39

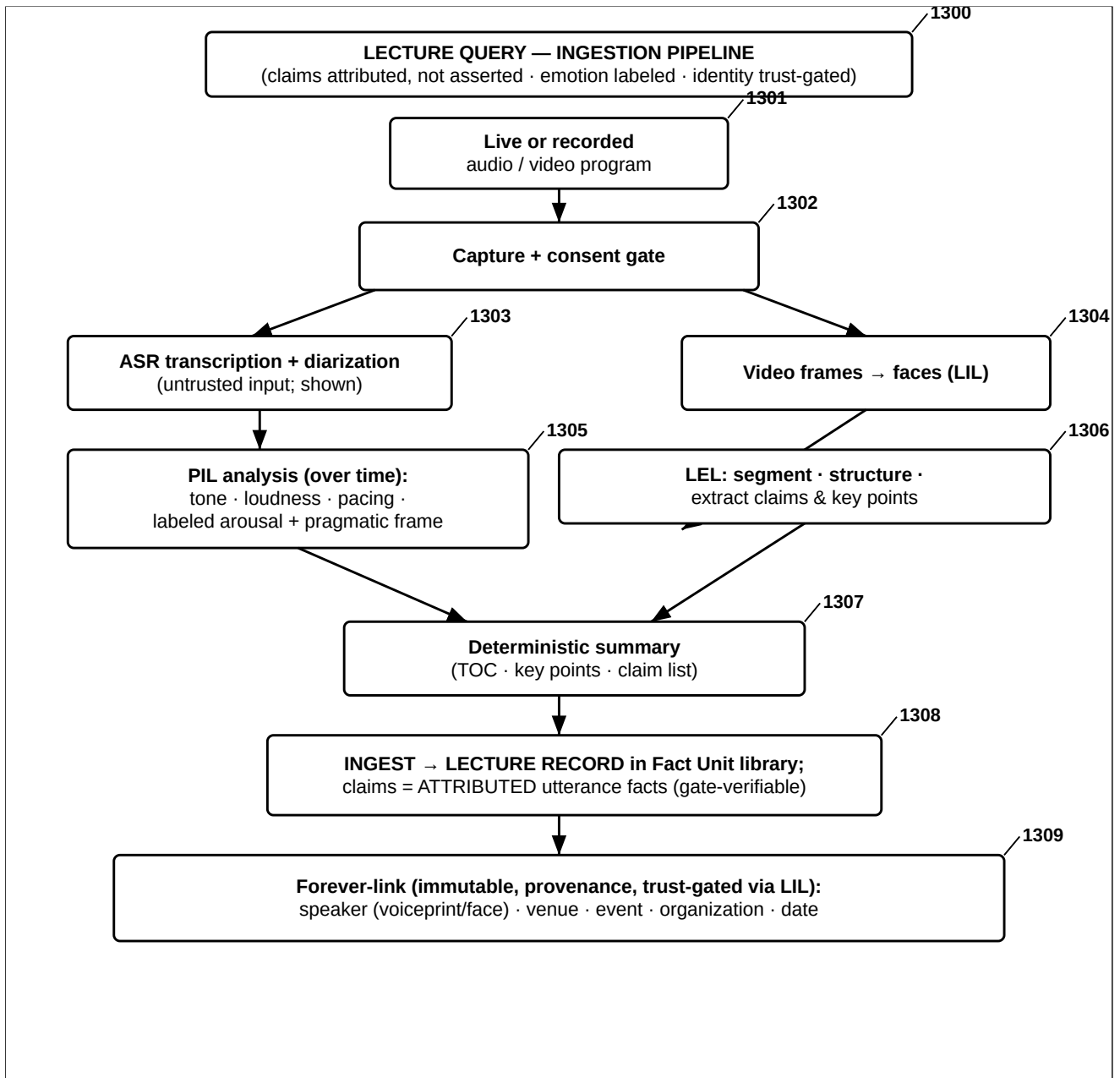


FIG. 39 — Lecture Query — Ingestion Pipeline

FIG. 40

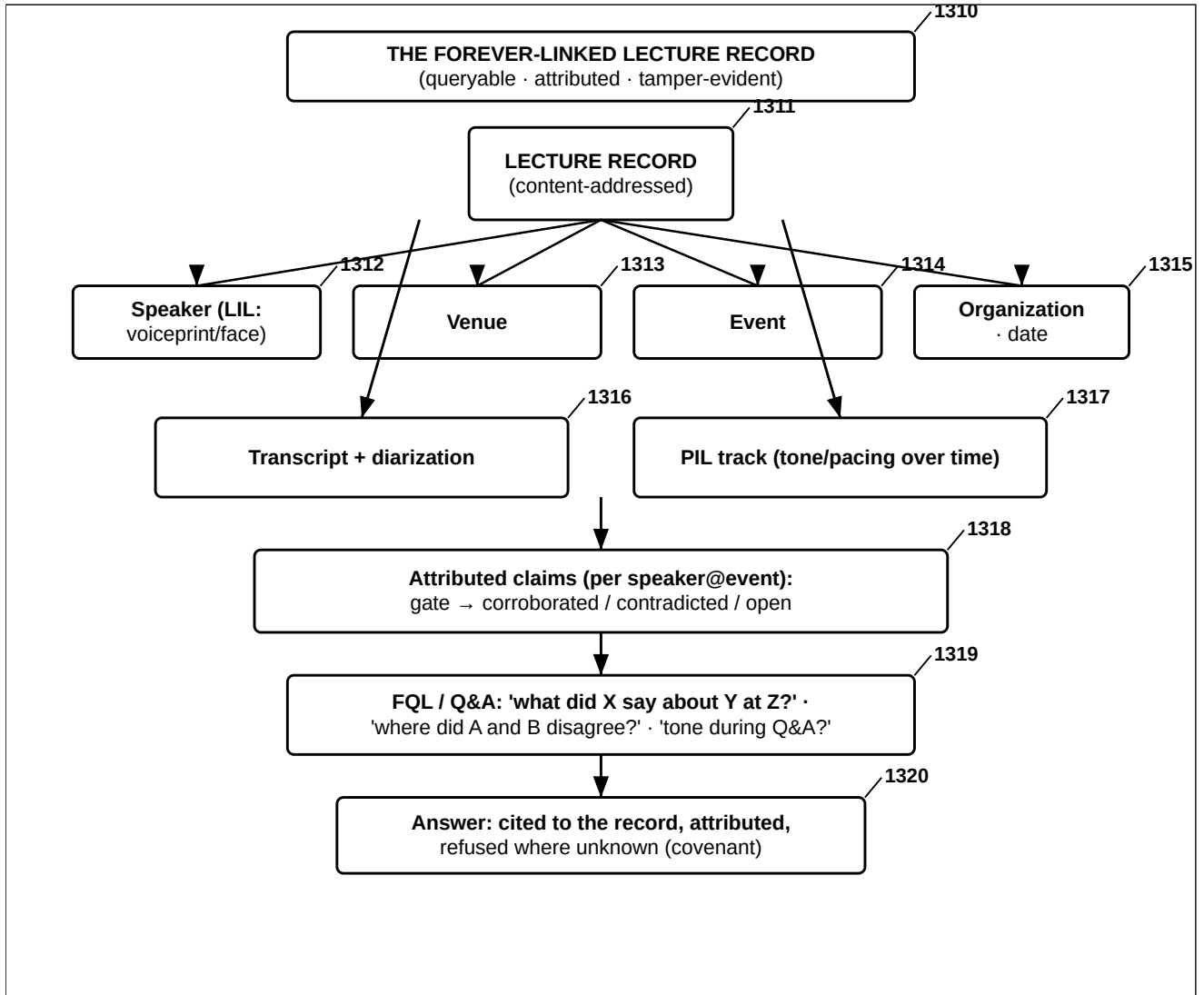


FIG. 40 — Lecture Query — The Forever-Linked Lecture Record and Its Querying

FIG. 41

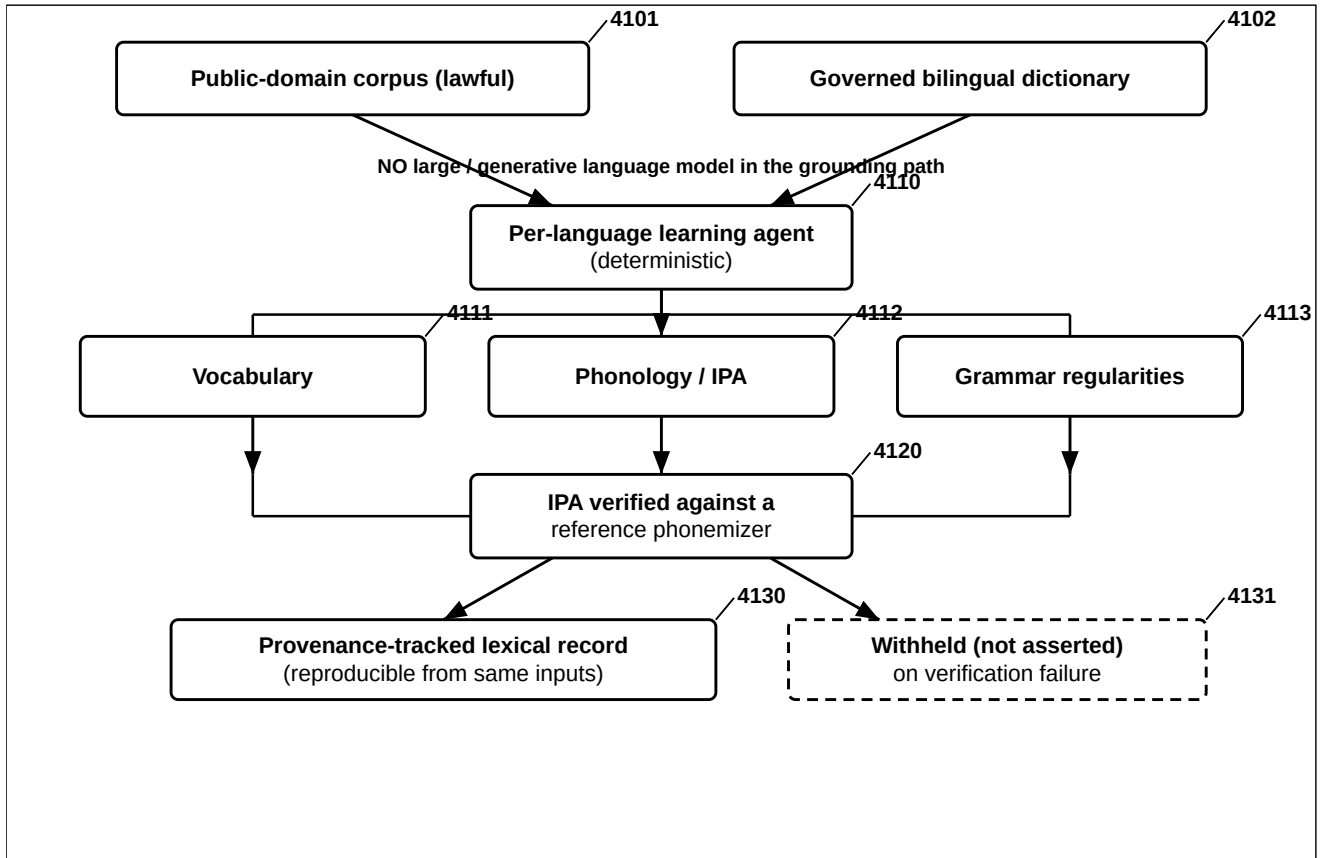


FIG. 41 — Deterministic Multilingual Acquisition Without a Generative Model

FIG. 42

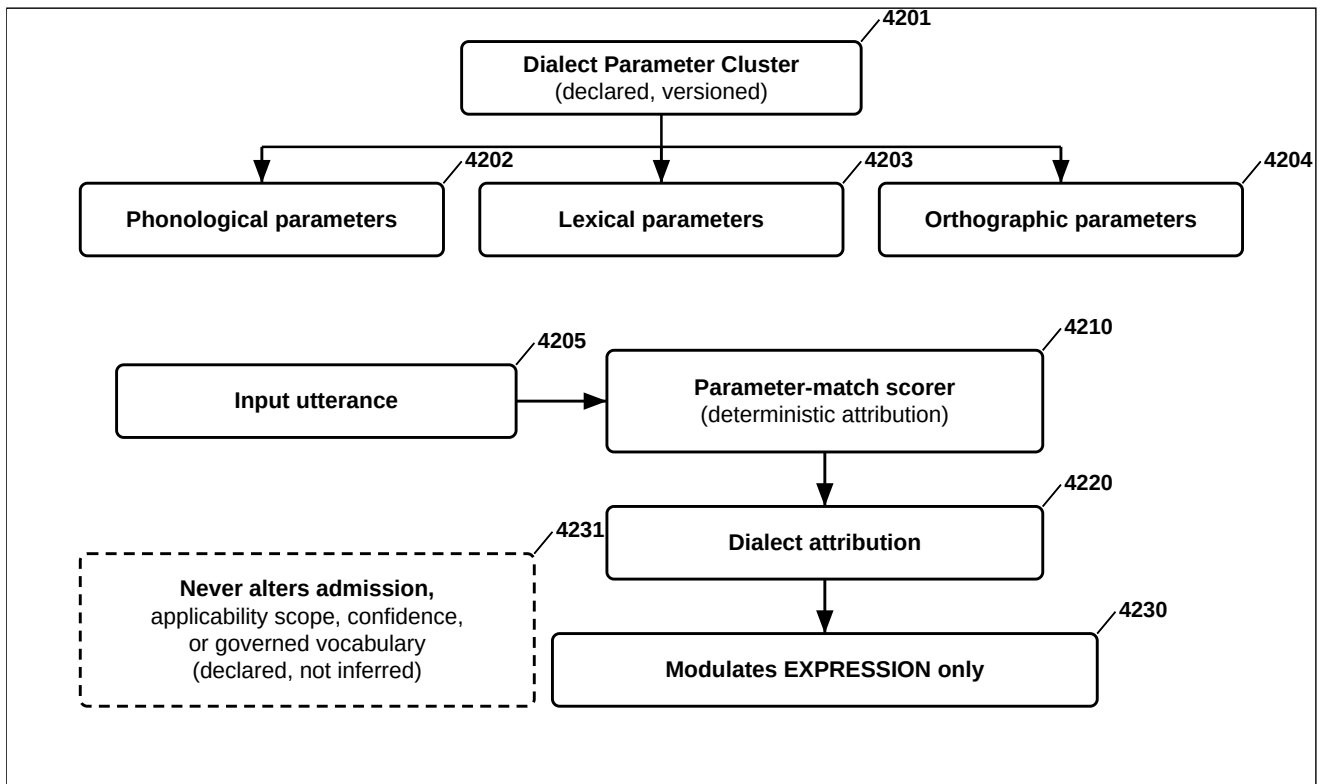


FIG. 42 — Dialect Parameter Clusters (DPC) — Declared, Expression-Only Modulation

FIG. 43

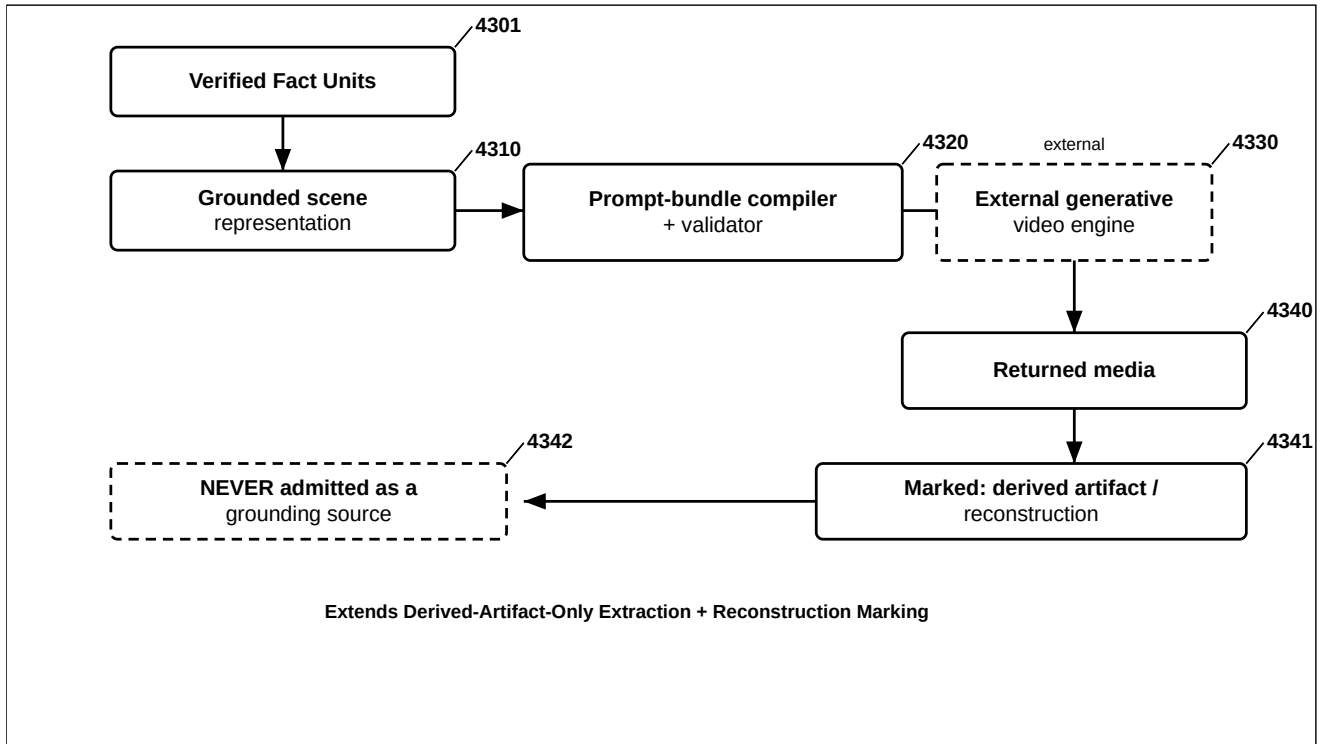


FIG. 43 — Grounded Scene Compilation to an External Generative Media Engine

FIG. 44

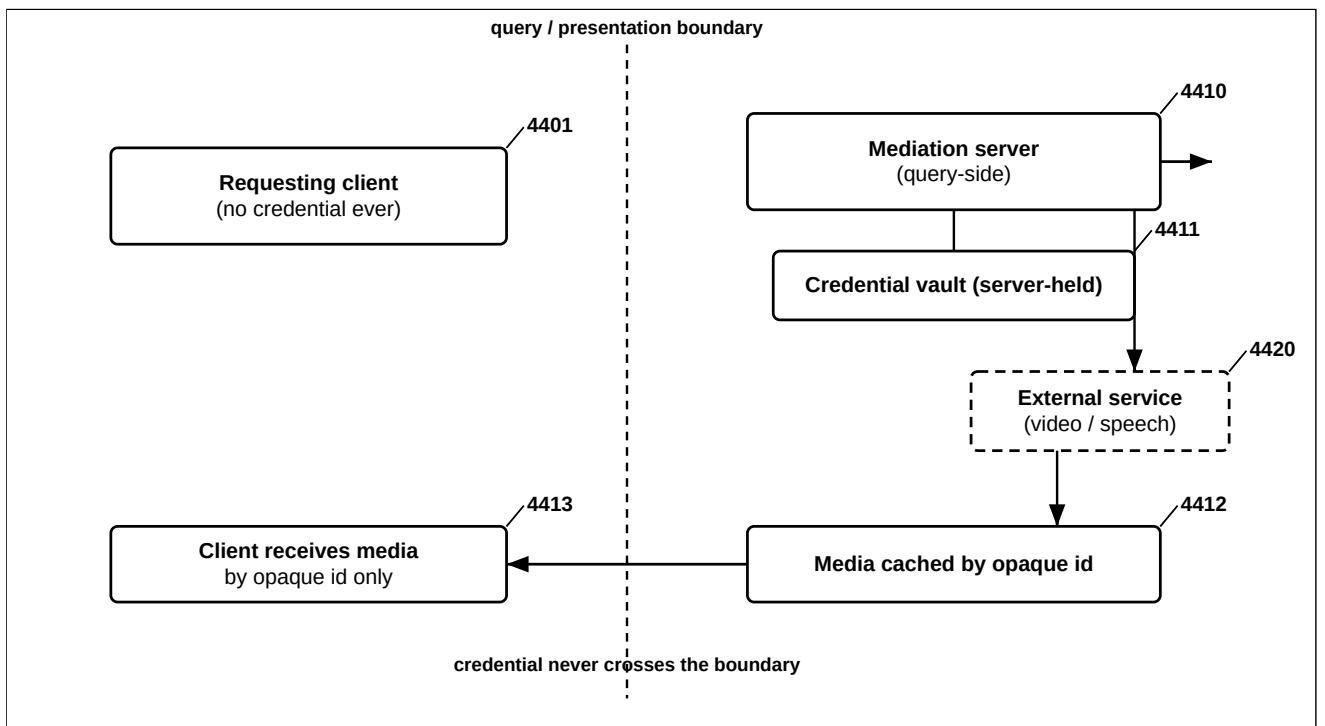


FIG. 44 — Key-Safe External-Service Mediation

FIG. 45

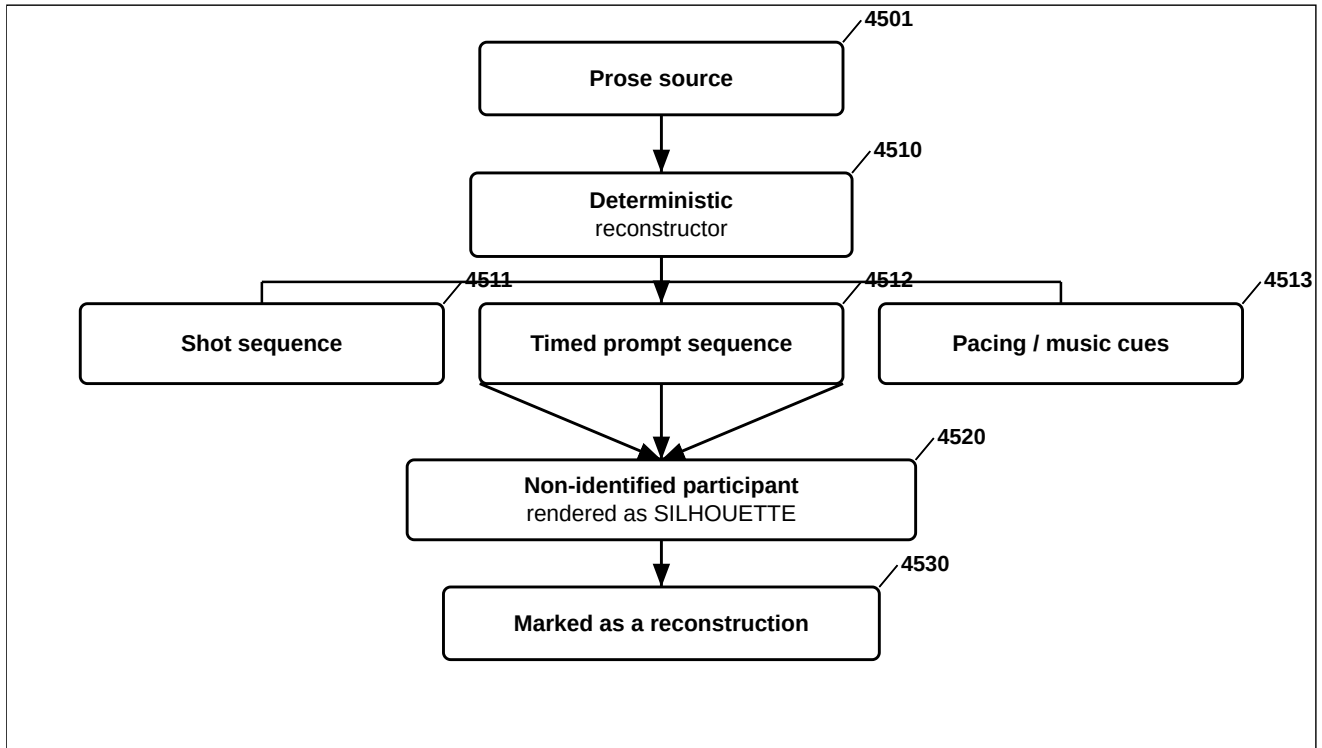


FIG. 45 — Scene Reconstruction From Prose — Bounded, Silhouette-Protected

FIG. 46

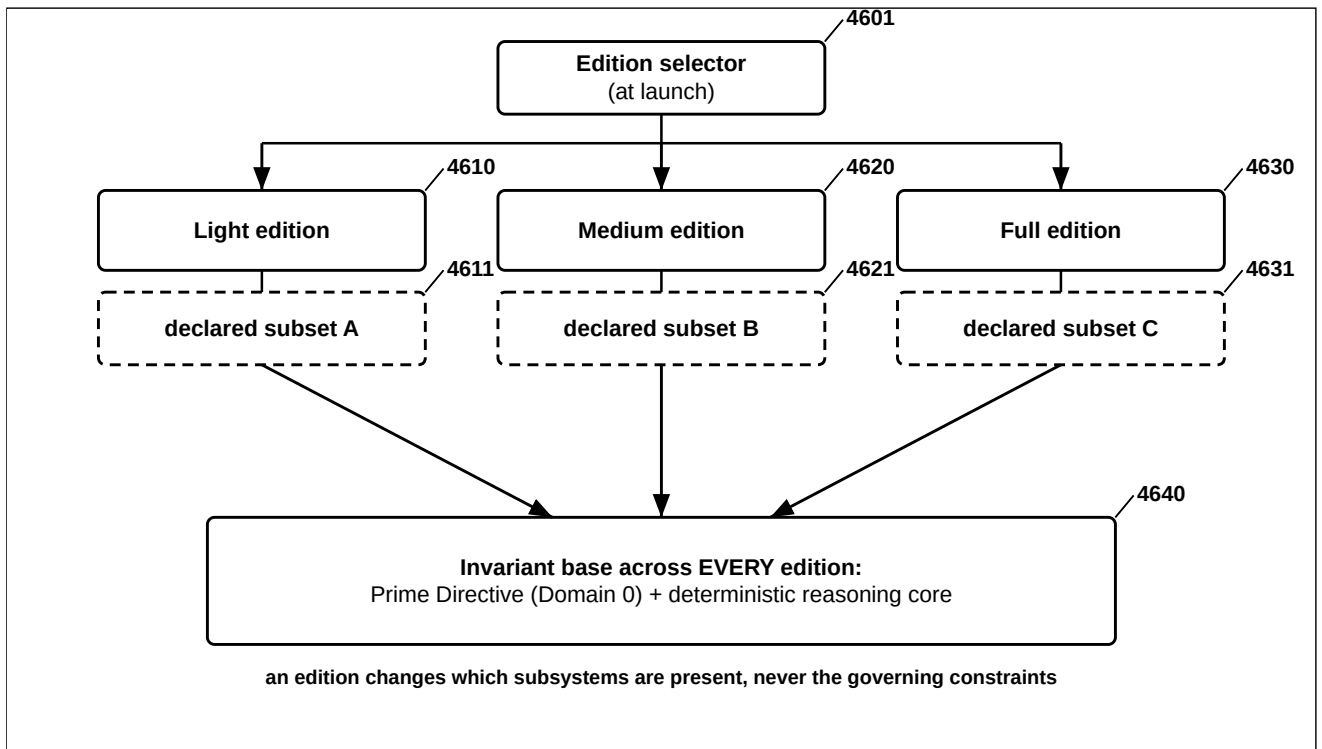


FIG. 46 — Modular Edition Packaging — Nested Capability Editions

FIG. 47

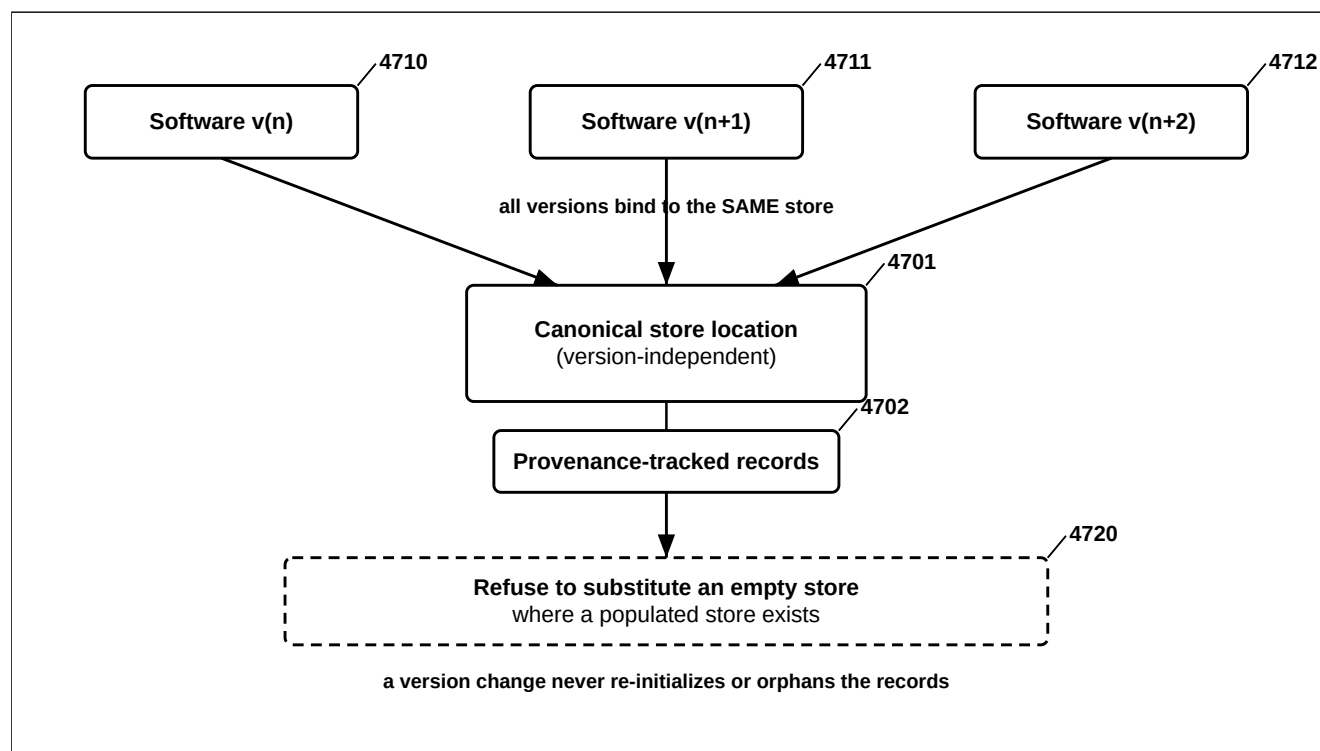


FIG. 47 — Version-Independent, Provenance-Continuous Store

FIG. 48

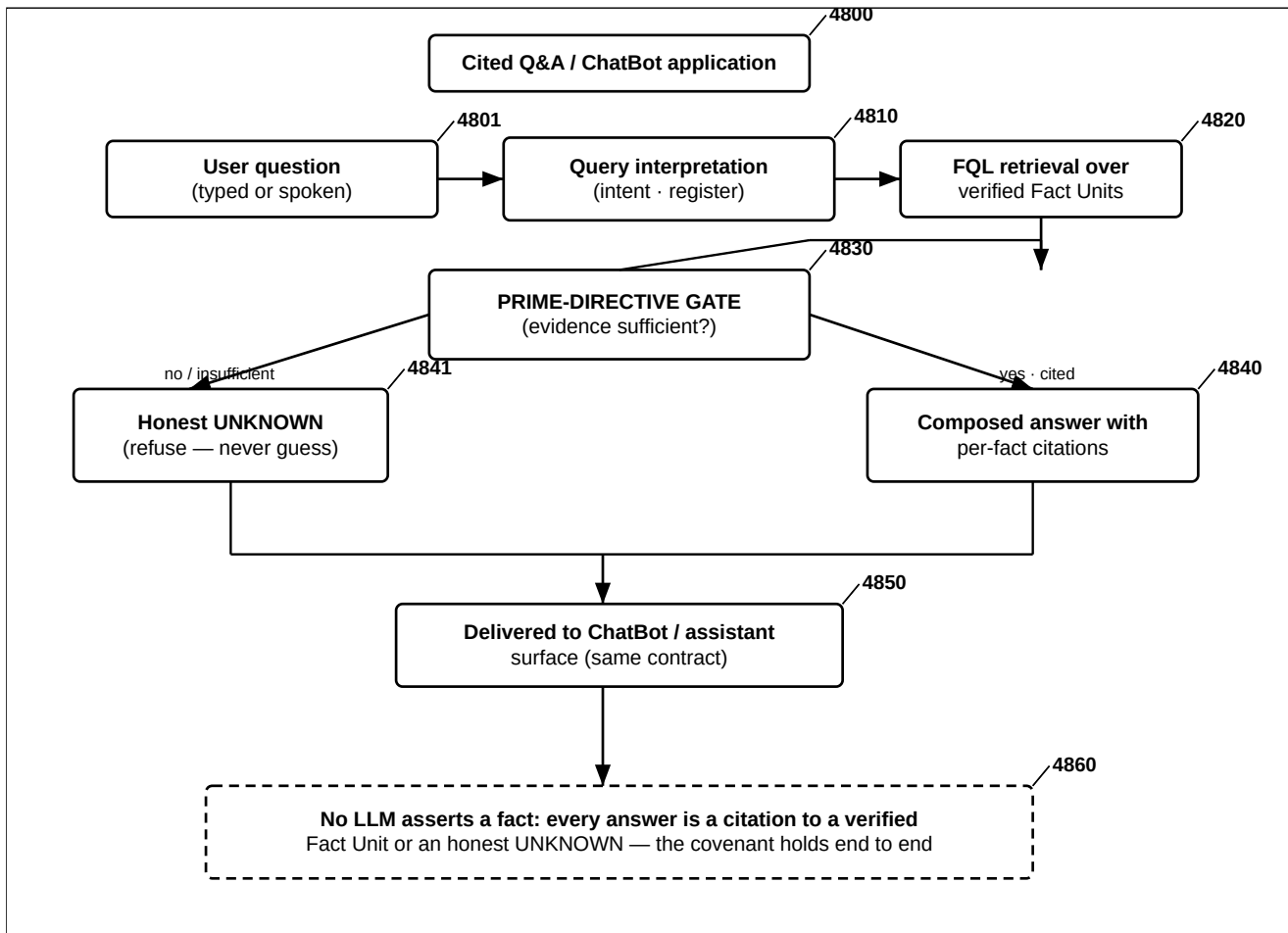


FIG. 48 — Cited Q&amp;A / ChatBot Application — Citation-or-UNKNOWN End to End

FIG. 49

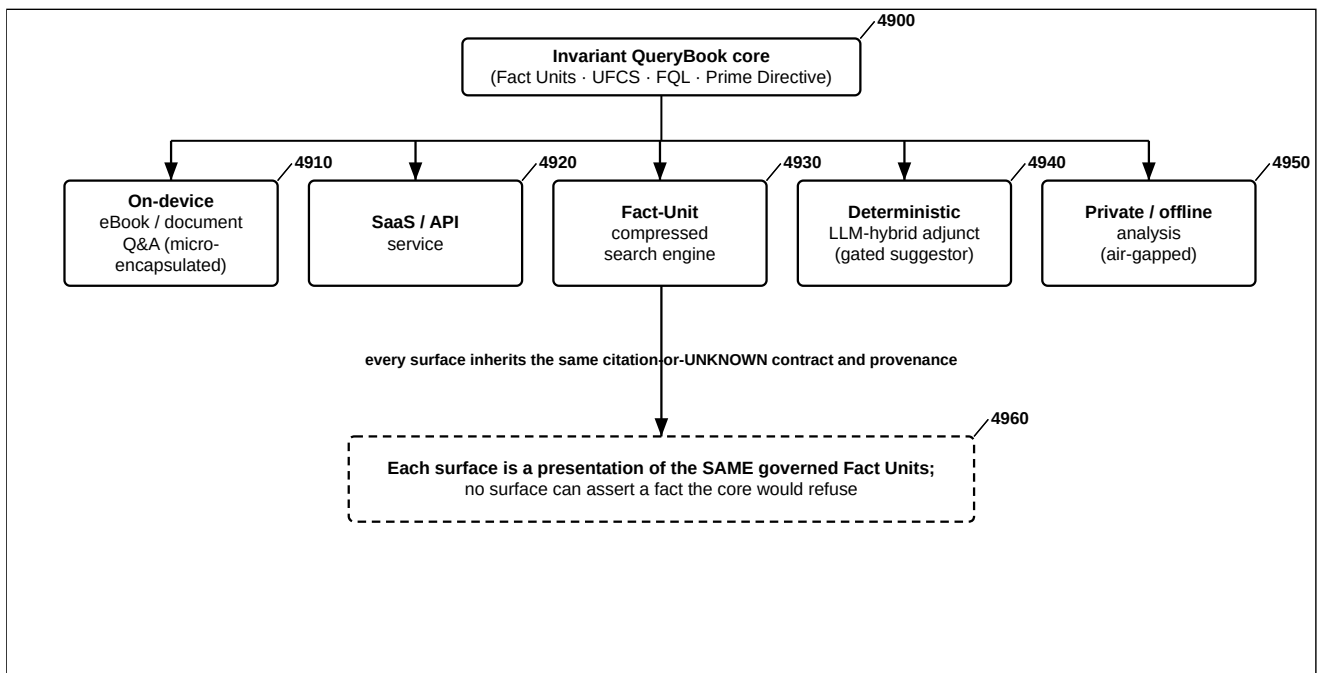


FIG. 49 — Application and Deployment Surfaces over the Invariant Core

DRAFT — NOT FILED — FOR REVIEW BY QUALIFIED PATENT COUNSEL PRIOR TO FILING. Companion drawings volume to the consolidated QueryBook Provisional Patent Application; figures and numerals are drawn from, and identical to, that application. Finalize to 37 CFR 1.84 before filing.